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BACHELOR THESIS

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Implicit Symplectic Integrators

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Title: Implicit Symplectic Integrators

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Abstract: This thesis studies symplectic integrators, a class of numerical solvers for ordinary differential equations which are suitable for solving problems from mechanics. The most important topics of numerical integration are presented, together with several formulations of classical mechanics. The most relevant is the Hamiltonian formulation and its connection to symplectic geometry. Afterwards, symplectic integrators are introduced together with several schemes and their important properties. Then, we theoretically investigate the perturbation of the symplectic structure matrix when implicit symplectic schemes are numerically solved; the results are somewhat novel. In particular, the Symplectic Euler scheme together with fixed point iteration method is studied. The error of symplecticity is estimated in case of one degree-of-freedom systems. The emergence of deviation terms in the matrix of symplectic structure is shown for larger systems. Finally, two problems from mechanics and electromagnetism are numerically solved using symplectic integrators. The second problem is to find magnetic field lines, which can be described using a nonstandard Hamiltonian.

Keywords: symplectic integrators, Hamiltonian systems, numerical integration, ordinary differential equations, magnetic field lines

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Abstrakt: Práce se zabývá symplektickými integrátory, které tvoří třídu numerických řešičů obyčejných diferenciálních rovnic vhodných pro řešení problémů mechaniky. Práce začíná shrnutím nejdůležitějších pojmů z oboru numerické integrace a přístupům k popisu klasické mechaniky. Významná je hamiltonovská formulace a její provázání se symplektickou geometrií. Práce zavádí symplektické integrátory, představuje některá schémata a jejich důležité vlastnosti. Následují vlastní teoretické výsledky studující poruchu matice symplektické struktury vlivem numerického řešení implicitních symplektických schémat. Konkrétně je uvažováno symplektické Eulerovo schéma v kombinaci s metodou pevného bodu. Chyba této metody je určena pro případ systému s jedním stupněm volnosti, u více stupňů volnosti je popsán vznik deviačních členů v matici symplektické struktury. Nakonec jsou pomocí symplektických integrátorů řešeny dvě úlohy na pomezí mechaniky a elektromagnetismu, přičemž druhá z nich se věnuje problému hledání magnetických indukčních čar, který lze popsat netradičním hamiltoniánem.

Klíčová slova: symplektické integrátory, hamiltonovské systémy, numerická integrace, obyčejné diferenciální rovnice, magnetické indukční čáry

Contents

Introduction

Differential equations are traditionally one of the most important tools used to mathematically model and analyse natural phenomena, be it motion, heat flow or populations in ecology. As these equations describe such a vast amount of problems, there is no unified method giving their exact solutions. In more complicated scenarios, it is often enough (and simpler) to find an approximate solution using a computer instead of trying to solve the original problem exactly. For that, many computational methods (numerical integrators) have been developed, carrying new questions about their performance, speed and robustness.

In this thesis, we present a particular class of such computational methods for solving ordinary differential equations, the so-called symplectic integrators. These integrators are specifically designed for Hamilton's equations, a type of ordinary differential equations which have a specific structure and properties, yet are quite abundant in physics (particularly in mechanics), and therefore important. Hamilton's equations admit several geometric invariants, one of them being a symplectic 2-form, and symplectic integrators respect this invariance, at least theoretically.

One of the aims of this thesis is to show that the symplectic invariance is not perfect even for symplectic integrators, as these methods are often implicit and have to be supplemented by a root-finding algorithm during actual computation, which is not assumed in the theoretical derivation of the symplectic form invariance. This nuance is often ignored in literature, yet it has a negative effect on the performance of these integrators. We present novel results about this symplectic error for a particular integrator (Symplectic Euler method) and root-finding algorithm (fixed point iteration method).

The thesis is structured in the following way: In Chapters 1 and 2, an introduction to ordinary differential equations and their numerical integrators, and to Hamilton's equations in mechanics and its underlying symplectic geometry, is presented. In Chapter 3, symplectic integrators are defined and some of their properties are discussed. These three chapters present an introductory background to the topic. In Chapter 4, we mathematically investigate how symplectic integrators behave when a root-finding algorithm has to be used in order to solve implicit relations given by the integrators. The results are compared with similar results in literature. Finally, in Chapter 5, we use symplectic integrators to numerically solve two problems. We also briefly present how electromagnetic field affects charged particles' motion and how the problem of finding magnetic field lines can be formulated using nonstandard Hamilton's equations, as those are the studied problems.

1 Numerical integration of ordinary differential equations

In this chapter, ordinary differential equations (ODEs) and their numerical solving methods are briefly presented as a minimal background for the symplectic integrators. For a much more complete discussion of these (standard and mature) mathematical topics, see e.g. Kurzweil's rigorous analytical textbook about ODEs [1], or Braun's textbook [2] and the textbook [3] of Hairer et al. for introduction to the theory of ODEs together with the numerical aspects of their solving.

1.1 Ordinary differential equations

By *numerical integration*, we mean numerical methods for solving ordinary differential equations. We restrict ourselves only to systems of first-order ordinary differential equations of the form

$$y' = f(t, y), \text{ for } t \in (a, b), \qquad y(t_0) = y_0, \tag{1.1}$$

where $a, b \in \mathbb{R}$, $a < b$, $t \in [a, b]$ is interpreted as time, $G \subset \mathbb{R}^N$ is an open set, $y : [a, b] \to G$ is the unknown function of t (continuously extended to the boundaries of $[a, b]$), $f : [a, b] \times G \to \mathbb{R}^N$ is given and $y_0 \in \mathbb{R}$ is a given *initial value* at the *initial time* $t_0 \in [a, b]$. Because this thesis studies symplectic integrators, which are used to solve Hamiltonian systems having this precise form, restriction to (1.1) does not cause any harm.

If f does not depend on t, the system (1.1) is called *autonomous*. The Hamiltonian systems considered in this work will usually have this property.

Due to the specification of $y(t_0)$, (1.1) is called an *initial value problem* (IVP). For f continuous on $[a, b] \times G$ and locally Lipschitz-continuous with respect to y, the Peano and Picard theorems guarantee that the problem (1.1) is locally well-posed for any initial values, i.e. that the solution exists and is unique on a neighbourhood of t_0.

From now on, without loss of generality, we set $t_0 = 0$ and work on the interval $[0, \tau]$ for some $\tau > 0$.

Suppose that (1.1) has a unique solution for every $y_0 \in G$ on the same interval $[0, \tau]$, $\tau > 0$ independent of y_0. For a particular choice of $y_0 = z$, we have a solution $y_z : [0, \tau] \to G$. However, we can instead fix $h \in [0, \tau]$ and define the *flow* of (1.1) as

$$\varphi_h : G \to G, \quad z \to y_z(h),$$

i.e. as a mapping that maps every initial value y_0 to the corresponding IVP solution evaluated at time h. This offers a different geometrical view on the solution, where instead of looking at the evolution of one particular initial value, we look how a larger set of initial values evolves after a certain time h.

1.2 Numerical integrators

The standard approach to solve (1.1) numerically on the time interval $[0,\tau]$ is to *discretise* the problem, i.e. divide this interval into subintervals $[t_i, t_{i+1}]$, $i \in \{0,\dots,n-1\}$, $t_0 = 0$, $t_n = \tau$, and then compute y_k as an approximation of the true value $y(t_k)$ using some recurrence relationships between evaluations of f, partition points t_i and previous approximations y_0, ..., y_{k-1} (where y_0 is the initial value from (1.1)). We shall call these recurrences a particular *numerical scheme*. It is often further assumed (as we also do from now on) that the interval partition is *equidistant*, i.e.

$$\forall i \in \{0,\dots n-1\} : t_{i+1} - t_i = h = \text{const}, \qquad h > 0.$$

The numerical scheme can depend on more previous approximations y_{k-1}, y_{k-2}, ..., y_{k-l} for some $l \geq 2$, in which case we say that the numerical scheme is *multi-step*. However, we shall deal only with *single-step* numerical schemes/methods in this work, which utilises only y_{k-1} for the calculation of y_k.

A particular single-step method can be formalised as

$$y_{k+1} = \Psi_h(y_k, y_{k+1}, t_k, t_{k+1}). \tag{1.2}$$

The function Ψ_h uniquely describes the single-step numerical scheme and we shall call it the *step function* of the scheme. We also call $\Phi_h : y_k \mapsto y_{k+1}$ the *numerical flow*, because it approximates the action of the ODE flow $\varphi_h : y(t_k) \mapsto y(t_{k+1})$.

Note that Φ_h and Ψ_h can differ, as Ψ_h can in general depend on y_{k+1} (so then Ψ_h gives the equation (1.2), whose solution defines Φ_h); in that case, the scheme is called *implicit*, otherwise it is called *explicit*. Explicit schemes are easier to analyse and implement and tend to be cheaper to evaluate, because implicit schemes usually yield nonlinear algebraic equations which need to be solved approximately using iterative methods, e.g. fixed point iteration or Newton's method. However, implicit schemes tend to be more stable, and furthermore in the setting of symplectic integrators, which this work studies, the schemes are almost always implicit.

Example. Using the Taylor polynomial and assuming $y \in \mathcal{C}^2([0,\tau])$, we can write

$$y(t_k + h) = y(t_k) + hy'(t_k) + \mathcal{O}(h^2) = y(t_k) + hf(t_k, y(t_k)) + \mathcal{O}(h^2), \quad h \to 0$$

and approximate

$$y_{k+1} = y_k + hf(t_k, y_k). \tag{1.3}$$

This is known as the *Explicit Euler method.* The step function is in this case

$$\Psi_h(y_k, y_{k+1}, t_k, t_{k+1}) = y_k + hf(t_k, y_k)$$

and is equal to the numerical flow Φ_h.

We can also proceed backwards and using

$$y(t_{k+1} - h) = y(t_{k+1}) - hy'(t_{k+1}) + \mathcal{O}(h^2) = y(t_{k+1}) - hf(t_{k+1}, y(t_{k+1})) + \mathcal{O}(h^2)$$

define the scheme

$$y_k = y_{k+1} - hf(t_{k+1}, y_{k+1}),$$

rewritten as

$$y_{k+1} = y_k + hf(t_{k+1}, y_{k+1}),$$

which is called *Implicit Euler method*. In this case, the step function Ψ_h is in general not equal to the numerical flow Φ_h.

Definition 1. Let $y : [0, \tau] \to \mathbb{R}^n$ be the solution of the initial value problem (1.1) and $(y_i)_{i=0}^n$ be the approximations of $(y(t_i))_{i=0}^n$ given by (1.2). The *global error* of the k-th step is defined as

$$e_k := y(t_k) - y_k.$$

The *local truncation* (or *discretisation*) *error* is defined as

$$\delta_{k+1} := y(t_{k+1}) - \Phi_h(y(t_k)),$$

The local truncation error measures the discrepancy between the true value of y at a time t_{k+1} and the numerical result y_{k+1} after a single step of the scheme given by Φ_h, provided that the previous approximation is as correct as it can be, thus setting it equal to the true value $y(t_k)$. The global error e_k takes into account all k steps of the scheme, whose individual errors can affect the other steps in nontrivial way.

Definition 2. Let Φ_h be the numerical flow of a particular numerical scheme. We say that this numerical scheme is of *order* $p \in (0, \infty)$, if for every IVP (1.1), it holds that

$$y(h) - \Phi_h(y_0) = \mathcal{O}(h^{p+1}), \quad h \to 0.$$

The order is given by the discrepancy between the exact ODE flow φ_h and the numerical flow Φ_h. It is often the case (under mild assumptions) that the global error of a p-order scheme is $\mathcal{O}(h^p)$. The standard analytical way to find a scheme's order is to expand $y(t_0 + h)$ into Taylor's polynomial, so that the terms of Φ_h cancel out with the first terms of the Taylor's polynomial. Here is an example:

Example. We will show that for $y \in \mathcal{C}^2([0, \tau])$, Explicit Euler method has order 1:

$$\begin{aligned}\delta_{k+1} = y(t_{k+1}) - (y(t_k) + hf(t_k, y(t_k))) = y(t_k) + hy'(t_k) + \mathcal{O}(h^2) - \\ - y(t_k) - hf(t_k, y(t_k)) = \mathcal{O}(h^2), \quad h \to 0,\end{aligned}$$

because $y'(t_k) = f(t_k, y(t_k))$.

1.2.1 Runge-Kutta Methods

The most commonly used numerical integrators are from the family of *Runge-Kutta (RK) methods*. Due to their general definition, these methods are quite flexible—in particular, there exist both explicit and implicit RK methods of arbitrary orders, and also some of the methods are symplectic, and thus relevant for this work.

Definition 3. Let $s \in \mathbb{N}$, $A = (a_{ij})_{i,j} \in \mathbb{R}^{s \times s}$ be a square matrix and $b, c \in \mathbb{R}^s$. These parameters define an *s-stage Runge-Kutta method*

$$\begin{aligned} k_i &= f(t_n + hc_i, y_n + h\sum_{j=1}^{s} a_{ij}k_j), \quad i \in \{1, \ldots, s\}, \\ y_{n+1} &= y_n + h\sum_{i=1}^{s} b_i k_i. \end{aligned} \tag{1.4}$$

The methods' parameters are usually described using the *Butcher's tableau*, visualised in Table 1.1.

$$
\begin{array}{c|cccc}
c_1 & a_{11} & a_{12} & \dots & a_{1s} \\
c_2 & a_{21} & a_{22} & \dots & a_{2s} \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
c_s & a_{s1} & a_{s2} & \dots & a_{ss} \\
\hline
 & b_1 & b_2 & \dots & b_s
\end{array}
$$

Table 1.1 Butcher's tableau for a general Runge-Kutta method.

The parameters a_{ij} are often chosen so that the individual k_i can be, if ordered correctly, computed explicitly, making the numerical scheme explicit—this is for example the case if A is lower or upper triangular with zero diagonal.

Moreover, the choice of the parameters A, b and c cannot be arbitrary—the method should be at least *consistent*, i.e. it should hold that the local discretisation error $\delta_n \to 0$ as $h \to 0$. Actually, the order of the method is determined by the parameters (for any order $p > 0$, the required consistency immediately follows) and there is a whole theory characterising the relationships between parameters and the order of the method or its other properties—this topic is however outside of the scope of this work.

Note that both Explicit and Implicit Euler methods are RK methods: For both methods, $s = 1$ and $b_1 = 1$; Explicit Euler uses $c_1 = a_{11} = 0$; Implicit Euler uses $c_1 = a_{11} = 1$. Another example is the widely used RK4, given by the Table 1.2.

$$
\begin{array}{c|cccc}
0 & 0 & 0 & 0 & 0 \\
1/2 & 1/2 & 0 & 0 & 0 \\
1/2 & 0 & 1/2 & 0 & 0 \\
1 & 0 & 0 & 1 & 0 \\
\hline
 & 1/6 & 1/3 & 1/3 & 1/6
\end{array}
$$

Table 1.2 Runge-Kutta 4.

For symplectic numerical integration, a somewhat more general class of numerical integrators is commonly used to solve the autonomous systems $y' = f(y)$, described in [3] and [4]. The function f is viewed as $f = (g_1, g_2)^T$ for g_1 having a subset of $\mathbb{R}^{m_1}$ as its range and g_2 having a subset of $\mathbb{R}^{m_2}$ as its range, where $m_1 + m_2 = N$. Then different RK parameters can be used for updating the first m_1 and the other m_2 components of the approximations y_n:

Definition 4. Let $s \in \mathbb{N}$, $A = (a_{ij})_{i,j}$, $\hat{A} = (\hat{a}_{ij})_{i,j}$ be two $s \times s$ square matrices and

$b, \hat{b} \in \mathbb{R}^s$. These parameters define an *s-stage partitioned Runge-Kutta method*

$$\begin{aligned} k_i &= g_1(u_n + h\sum_{j=1}^{s} a_{ij}k_j, v_n + h\sum_{j=1}^{s} \hat{a}_{ij}\ell_j), \quad i \in \{1, \dots, s\}, \\ \ell_i &= g_2(u_n + h\sum_{j=1}^{s} a_{ij}k_j, v_n + h\sum_{j=1}^{s} \hat{a}_{ij}\ell_j), \quad i \in \{1, \dots, s\}, \\ u_{n+1} &= u_n + h\sum_{i=1}^{s} b_i k_i, \\ v_{n+1} &= v_n + h\sum_{i=1}^{s} \hat{b}_i \ell_i. \end{aligned} \tag{1.5}$$

1.2.2 Backward error analysis

The local truncation error and the global error are the main tools to study the *forward error* of numerical integrators, i.e. estimate the distance between the exact and the approximate solution of the ODE. However, there is an alternative approach, more commonly used in numerical linear algebra—*backward error analysis*. The idea is to find and analyse a *modified equation* which the integrator actually solved exactly. Thus, instead of comparing the true and approximate solutions, as in forward error analysis, the true and modified problems (ODEs) are compared, justifying the name backward error analysis. We briefly introduce the idea and a few of its results without proofs, as this thesis is more interested in applications of this backward analysis approach for symplectic integrators, and that topic will be presented more thoroughly later on. For reference, see e.g. [4] or [5].

Finding the modified equation is not simple and it requires quite strict assumptions. For simplicity, let the original equation be $y' = f(y)$ (f independent on t). The standard approach we employ here is to presume that the modified equation has the form

$$y' = f_0(y) + hf_1(y) + h^2 f_2(y) + \dots, \tag{1.6}$$

such that $y(ih) = y_i$, $i \in \{0, 1, \dots, n\}$, i.e. the solution of (1.6) in the partition points is exactly the numerical solution $(y_i)_{i=0}^n$. If the numerical method is consistent, it should hold that $f_0 = f$ (so the equation (1.6) reduces to the original problem as $h \to 0$).

Assume that the numerical flow Φ_h can be expanded as

$$\Phi_h(y) = \eta_0(y) + h\eta_1(y) + h^2\eta_2(y) + \dots$$

with η_i known beforehand for the particular scheme. Also assume that the solution of (1.6) can be expanded into Taylor series

$$y(t+h) = y(t) + hy'(t) + \frac{h^2}{2}y''(t) + \dots,$$

where the derivatives can be computed (for smooth f_i) from (1.6) using the chain rule, e.g.

$$\begin{aligned} y' &= f_0(y) + hf_1(y) + h^2 f_2(y) + \dots, \\ y'' &= (f_0'(y) + hf_1'(y) + h^2 f_2'(y) + \dots)(f_0(y) + hf_1(y) + h^2 f_2(y) + \dots) \end{aligned}$$

with successively more and more complicated expressions.

In order to have the solution of (1.6) equal to the numerical solution, we require $\Phi_h(y(t)) = y(t+h)$. From this expression, the unknown functions f_i can be in principle regained by comparing the coefficients of the same powers of h:

$$\eta_0(y) + h\eta_1(y) + h^2\eta_2(y) + \cdots = y + hf_0(y) + h^2(f_1(y) + \frac{1}{2}f_0'(y)f_0(y)) + \ldots$$

For example, we see that $f_1(y) = \eta_2(y) - \frac{1}{2}f'(y)f(y)$ (using the fact $f_0 = f$ for consistent method).

The properties of the numerical scheme are often reflected in the modified equation (See [4], Chapter IX, Theorem 1.2.):

Theorem 1. *For an order p numerical scheme, its modified equation (1.6) has the form*

$$y' = f(y) + h^p f_p(y) + h^{p+1} f_{p+1}(y) + \ldots.$$

The assumption on the existence of the Taylor expansion of Φ_h can be checked for the particular scheme. More problematic are the assumptions on the smoothness of the modified equation's solution y and on the form of the modified equation (1.6) itself—the series are not guaranteed to converge (and there are examples showing that the modified equation's series do diverge for particular choices of f and Φ_h).

The equation (1.6) thus usually has to be truncated, keeping only the first K terms of the right hand side. The numerical solution is then no longer an exact solution of this truncated modified ODE. However, because K can be chosen arbitrarily, the truncated ODE can still give us useful information about the numerical solution, as we shall see later on.

For the *K-truncated modified equation*

$$y' = f(y) + hf_1(y) + \cdots + h^K f_K(y), \tag{1.7}$$

let $\varphi_t^{[K]}$ be its ODE flow. The discrepancy between this flow and the numerical flow Φ_h, which formally "solved" the modified equation before its truncation, is (see [6], Theorem 4.1)

$$\Phi_h = \varphi_h^{[K]} + h^{K+1} f_{K+1} + \mathcal{O}(h^{K+2}), \quad h \to 0. \tag{1.8}$$

2 Hamiltonian systems

In this chapter, three important descriptions of classical mechanics (the study of motions) are introduced, with an emphasis on the Hamiltonian formalism. A brief introduction to symplectic geometry and its connection to Hamiltonian formalism will be also presented later on. For reference to analytical mechanics, see Goldstein's [7], Arnold's [8] and Podolský's [9] physics textbooks, or the textbook [4] (Chapter VI), giving broader context for numerical integration of physical systems.

We slightly adjust the notation in this chapter in order to mimic how physics equations are usually written. Vector quantities $\boldsymbol{r} \in \mathbb{R}^N$ shall be bold. The derivative with respect to time is denoted by a dot, i.e.

$$\dot{x}(t) := \frac{\mathrm{d}x}{\mathrm{d}t}(t) \qquad \ddot{x}(t) := \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}(t).$$

In Lagrangian and Hamiltonian formalisms, it is customary to denote the individual generalised positions by superscripts instead of subscripts, i.e. by quantities q^1, $\ldots$, q^N instead of q_1, $\ldots$, q_N.

2.1 Formulations of classical mechanics

The first exposure to mechanics is usually done in the setting of *Newtonian formulation.* There are objects whose motions we study (usually in the Euclidean space $\mathbb{R}^3$) and *forces* $\boldsymbol{F}_1, \ldots, \boldsymbol{F}_n$ which specify how the motions evolve. These forces are given by constraints of the motion and other interactions which stem from other physical theories.

The motion of a single *point particle* (an idealised dimensionless object, whose position in three-dimensional space is fully determined by a position vector $\boldsymbol{r}$ in a given Cartesian frame) subject to forces $\boldsymbol{F}_1, \ldots, \boldsymbol{F}_n$ is fully specified by the relation

$$m\ddot{\boldsymbol{r}} = \sum_{i=1}^{n} \boldsymbol{F}_i, \tag{2.1}$$

where $m > 0$ is the *mass* of the particle, a constant of proportionality in the equation and a property of the point particle. This is known as the *Newton's second law of motion.*

The relation (2.1) is a vector equation, which means that in a particular cartesian coordinate frame, it is a system of three second order ordinary differential equations. In order to construct this system for a particular problem, the geometry of the forces needs to be carefully taken into account and the vector character makes the description in non-cartesian coordinate systems (in *curvilinear systems* like cylindrical or spherical coordinates) more complicated.

Therefore, even though equation (2.1) is sufficient for description of mechanical systems, it might be convenient to use different formalisms of mechanics, which we shall now explore.

2.1.1 Lagrangian formalism

The first example is the *Lagrangian formalism*. Assume that the studied mechanical system can be uniquely described by *generalised coordinates* $q^1, \dots, q^N$—in three-dimensional space, these can be just three cartesian coordinates (x, y, z) of a chosen coordinate frame, but also cylindrical coordinates (ρ, θ, z) or some other coordinates. Then, the equations of motion are usually written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{\partial \mathcal{L}}{\partial \dot{q}^j}\Big) - \frac{\partial \mathcal{L}}{\partial q^j} = 0, \qquad j \in \{1, \dots, N\}, \tag{2.2}$$

where $\mathcal{L}$ is the *Lagrangian*, a smooth-enough function defined in (formally independent) variables $q^1, \dots, q^N$, $\dot{q}^1, \dots, \dot{q}^N$, t. More specifically, the partial derivatives are computed as if the variables of $\mathcal{L}$ were not anyhow related, which yields functions

$$\mu_j(t, q^i, \dot{q}^i) = \frac{\partial \mathcal{L}}{\partial \dot{q}^j}(t, q^i, \dot{q}^i), \qquad \nu_j(t, q^i, \dot{q}^i) = \frac{\partial \mathcal{L}}{\partial q^j}(t, q^i, \dot{q}^i);$$

then, the rewritten equation (2.2)

$$\frac{\mathrm{d}\mu_j}{\mathrm{d}t}(t, q^i(t), \dot{q}^i(t)) - \nu_j(t, q^i(t), \dot{q}^i(t)) = 0$$

is understood as a differential equation in independent variable t and dependent variables q^i (and thus q^i and $\dot{q}^i$ are again directly related via $\dot{q}^i(t) = \frac{\mathrm{d}q^i}{\mathrm{d}t}(t)$). The leftmost time derivative causes the ODE to be in general of second order. Because j goes from 1 to N, the motion is described by N second order ODEs in the Lagrangian formalism.

The function $\mathcal{L}$ plays the same role as forces $\boldsymbol{F}$ in Newtonian formalism, as it describes interactions in the studied problem and dictates the motion. However, $\mathcal{L}$ is a scalar function, which makes the computations and derivations of equations of motion simpler.

For mechanical systems, the Lagrangian is usually given as

$$\mathcal{L} = T - V,$$

where T is the total kinetic energy of individual particles and V are summed potential energies describing mechanical interactions (i.e. Newtonian gravity or interactions due to springs). However, more general settings admit the Lagrangian formulation, provided that the correct Lagrangian is chosen—as an example, a particle with electric charge Q subject to an external magnetic field given by vector potential $\boldsymbol{A}$ and electric field given by scalar potential ϕ has Lagrangian

$$\mathcal{L}_{\text{elmg}} = \frac{1}{2}mv^2 + Q\boldsymbol{v} \cdot \boldsymbol{A} - Q\phi,$$

where m is the particle's mass and $\boldsymbol{v}$ is its velocity. We shall discuss electromagnetic fields more thoroughly in Chapter 5 while numerically solving relevant problems.

2.1.2 Hamiltonian formalism

The central formalism in this work is the *Hamiltonian formalism*. Similarly to the Lagrangian formalism, the interactions are described by a single scalar function, which is called the *Hamiltonian* and denoted as $\mathcal{H}$. The Hamiltonian is a smooth-enough function of t, $q^1, \dots, q^N$ and $p_1, \dots, p_N$. Here, q^i are again generalised coordinates, but instead of variables $\dot{q}^i$ (as it was the case in the Lagrangian formalism), the rest of variables of $\mathcal{H}$ are *canonical momenta* $p_1, \dots, p_N$. Together, coordinates q^i and p_i describe a position in an abstract *phase space*.

The relationship between generalised coordinates q^i and momenta p_i is a bit nuanced. Formally, when the Hamiltonian is already computed, these variables are all independent. However, when deriving the Hamiltonian from the Lagrangian, the individual positions and momenta are closely related.

Suppose we have a Lagrangian $\mathcal{L}$ with generalised coordinates $q^1, \dots, q^N$. Then, for particular $j \in \{1, \dots, N\}$, the *conjugate momentum to* q^j is defined as

$$p_j := \frac{\partial \mathcal{L}}{\partial \dot{q}^j}$$

(where the variables t, q^i and $\dot{q}^i$ of $\mathcal{L}$ are again understood as formally independent symbols).

This describes N implicit relations between t, q^i, $\dot{q}^i$ and p_i. If (at least locally) these relations yield functions[1] $\dot{q}^j(t, q^i, p_i)$, then the Hamiltonian is derived from the Lagrangian as

$$\mathcal{H}(t, q^i, p_i) = \sum_{k=1}^{N} p_k \dot{q}^k(t, q^i, p_i) - \mathcal{L}(t, q^j, \dot{q}^j(t, q^i, p_i)).$$

In the Hamiltonian formalism, the equations of motion, called *Hamilton's canonical equations*, are

$$\dot{q}^j = \frac{\partial \mathcal{H}}{\partial p_j}(t, q^i, p_i), \qquad \dot{p}_j = -\frac{\partial \mathcal{H}}{\partial q^j}(t, q^i, p_i), \qquad j \in \{1, \dots, N\}, \tag{2.3}$$

which are $2N$ first order ODEs in independent variable t and dependent variables q^i and p_i. Moreover, this system of ODEs is already of the form (1.1).

2.2 Properties of the Hamiltonian formalism

We begin this section by introducing useful invariants of motion.

Definition 5. Let $\mathcal{H}$ be a Hamiltonian on $[0, \tau] \times G$ for $G \subset \mathbb{R}^{2N}$ open, and $I : G \to \mathbb{R}$. We say that I is an *integral of motion* given by $\mathcal{H}$, if, for every solution (q, p) of the system (2.3) in G, the function I is constant along this solution, i.e. if it holds that

$$\forall t \in [0, \tau] : I(q(t), p(t)) = I(q(0), p(0)).$$

[1] For example by explicitly computing it or by using the implicit function theorem.

Integrals of motion are useful, because they impose additional constraints on the individual unknown functions q^i, p_i, which often leads to not having to solve the whole system (2.3). Note that the previous definition does not have to be specific for the Hamiltonian formalism, as the solutions/trajectories could be regained from any other mechanical formalism. However, a remarkable property of the Hamiltonian formalism, which in general does not hold for forces or Lagrangians, is this theorem:

Theorem 2. *Let $\mathcal{H}$ be a $\mathcal{C}^1$ Hamiltonian independent of time, i.e. $\mathcal{H}$ is a function only of position and momenta coordinates from $G \subset \mathbb{R}^{2N}$. Then, $\mathcal{H}$ is an integral of motion given by $\mathcal{H}$.*

Proof. Let q^j, p_j be a solution of the system (2.3) on time interval $[0, \tau]$. Then, for $t \in (0, \tau)$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}(q^j(t), p_j(t)) &= \sum_{i=1}^{n} \Big(\frac{\partial \mathcal{H}}{\partial q^i}(q^j(t), p_j(t)) \cdot \dot{q}^i(t) + \frac{\partial \mathcal{H}}{\partial p_i}(q^j(t), p_j(t)) \cdot \dot{p}_i(t) \Big) = \\ &= \sum_{i=1}^{n} (-\dot{p}_i(t)\dot{q}^i(t) + \dot{q}^i(t)\dot{p}_i(t)) = 0,\end{aligned}$$

where we used the chain rule and the equations (2.3), which q^i, p_i satisfy. Therefore, $\mathcal{H}(q^j(\cdot), p_j(\cdot))$ is a constant function on $[0, \tau]$, as differentiable $\mathcal{H}$ is continuous on G and q^j, p_j are continuous on $[0, \tau]$ as ODE solutions. □

If $\mathcal{H}$ is an integral of motion, we shall refer to it as (*generalised*) *energy*. This is motivated by the fact that for many mechanical systems, the Hamiltonian is given by $\mathcal{H} = T + V$, i.e. the total energy of the system. In such cases, the kinetic energy T is usually independent of positions q^i and the potential energy V independent of momenta p_i, which then makes $\mathcal{H}$ separable according to this definition:

Definition 6. A Hamiltonian $\mathcal{H}$ is said to be *separable* if it can be written as the sum $\mathcal{H}(t, q^i, p_i) = Q(t, q^i) + P(t, p_i)$ for some functions Q, P. Otherwise, $\mathcal{H}$ is called *nonseparable*.

We shall see in the following chapters that separable Hamiltonians are simpler to numerically integrate using symplectic integrators.

2.2.1 Symplectic structure

The Hamiltonian formalism can be formulated in a coordinate-free way using the language of symplectic geometry, a branch of differential geometry on manifolds. Let us briefly introduce some of the concepts of this geometry. We again utilise the sources [4] (Chapter VI), [8] (Chapters 8 and 9) and [9] (Book II, Chapter 3), and also the first chapters of the symplectic geometry textbook [10].

Symplectic structures and symplectomorphisms

Definition 7. Let V be a real vector space. We say that a bilinear form

$$\omega : V \times V \to \mathbb{R}$$

is a *symplectic structure* on V, if ω is:

- skew-symmetric: $\omega(u, v) = -\omega(v, u)$ for all $u, v \in V$;
- non-degenerate: if $u \in V$ such that $\omega(u, \cdot)$ is a zero mapping, then $u = 0$.

Definition 8. Let V and W be real vector spaces equipped with symplectic structures ω, σ, respectively. Let $\phi : V \to W$ be a linear isomorphism. We say that ϕ is a *symplectomorphism*, if for every $u, v \in V$, it holds that

$$\sigma(\phi(u), \phi(v)) = \omega(u, v).$$

Definition 9. The *canonical symplectic structure* on $\mathbb{R}^{2N}$ is defined as

$$\Omega(u, v) := u^T J v, \quad u, v \in \mathbb{R}^{2N},$$

where J is the *matrix of symplectic structure*

$$J := \begin{pmatrix} O_N & I_N \\ -I_N & O_N \end{pmatrix},$$

where $I_N, O_N \in \mathbb{R}^{N \times N}$ are identity and zero matrices, respectively.

Note that Ω is bilinear, it is skew-symmetric (because J is a skew-symmetric matrix, i.e. $J^T = -J$) and it is non-degenerate (because J is nonsingular—it is orthogonal, so it also holds that $J^{-1} = J^T$), so it is a symplectic structure.

The geometrical intuition for Ω is the following: For two vectors $u, v \in \mathbb{R}^{2N}$, the space understood as $\mathbb{R}^N \times \mathbb{R}^N$, we have

$$\Omega(u, v) = u^T \begin{pmatrix} O_N & I_N \\ -I_N & O_N \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_N \\ \hat{v}_1 \\ \vdots \\ \hat{v}_N \end{pmatrix} = \begin{pmatrix} u_1 & \cdots & u_N & \hat{u}_1 & \cdots & \hat{u}_N \end{pmatrix} \begin{pmatrix} \hat{v}_1 \\ \vdots \\ \hat{v}_N \\ -v_1 \\ \vdots \\ -v_N \end{pmatrix} =$$

$$= \sum_{i=1}^{N} (u_i \hat{v}_i - \hat{u}_i v_i) = \sum_{i=1}^{N} \det \begin{pmatrix} u_i & v_i \\ \hat{u}_i & \hat{v}_i \end{pmatrix},$$

which can be understood as projecting the vectors u, v to subspaces $\operatorname{span}\{e_i, e_{N+i}\}$, $i \in \{1, \ldots, N\}$, giving always projections $(u_i, \hat{u}_i)^T$ and $(v_i, \hat{v}_i)^T$, measuring oriented areas of parallelograms given by these projections, and finally summing all those areas together. In particular, when $N = 1$, $\Omega(u, v)$ is exactly the oriented area of a parallelogram given by the vectors u and v.

A linear isomorphism acting on $\mathbb{R}^{2N}$ can be represented by a nonsingular matrix $S \in \mathbb{R}^{2N \times 2N}$. The symplectomorphism condition then states that

$$u^T S^T J S v = \Omega(Su, Sv) = \Omega(u, v) = u^T J v.$$

Therefore, S defines a symplectomorphism on $\mathbb{R}^{2N}$ if and only if $S^TJS = J$. If this condition holds, we call S a *symplectic matrix*.

Symplectic matrices form a subgroup of the general linear group $\mathrm{GL}(2N)$:

1. The identity matrix I is symplectic.
2. If S and P are symplectic, then SP is symplectic:
$$(SP)^TJ(SP) = P^TS^TJSP = P^TJP = J.$$
3. If S is symplectic, then S^{-1} is symplectic: First, note that S^T is symplectic, because
$$SJS^T = SJS^TJS(JS)^{-1} = SJJ(JS)^{-1} = S(-I)S^{-1}J^{-1} = -IJ^{-1} = J,$$
using the fact that $J^2 = -I$. From that,
$$(S^{-1})^TJS^{-1} = (SJ^TS^T)^{-1} = ((SJS^T)^T)^{-1} = (J^T)^{-1} = J.$$

A *symplectic form* is a type of differential 2-form ω on a manifold $\mathcal{M}$. A differential 2-form ω can be thought of as an in some sense smooth mapping which maps a point $x \in \mathcal{M}$ to a skew-symmetric bilinear form ω_x on the tangent space $T_x\mathcal{M}$ of the manifold $\mathcal{M}$ in the point x. A 2-form ω is a symplectic form, if it is *non-degenerate* (ω_x is non-degenerate in every $x \in \mathcal{M}$) and *closed* ($\mathrm{d}\omega = 0$, where d is the exterior derivative). For $x \in \mathcal{M}$, ω_x is already skew-symmetric bilinear form on $T_x\mathcal{M}$, so the non-degenerate condition for ω ensures that ω_x is a symplectic structure on $T_x\mathcal{M}$. If ω is a symplectic form on $\mathcal{M}$, we call $\mathcal{M}$ a *symplectic manifold*.

The reason why we do not go into the details of general symplectic geometry is the *Darboux theorem*. It informally states that a symplectic manifold is locally equivalent to $\mathbb{R}^{2N}$ equipped with the canonical symplectic structure Ω (more precisely, we should equip $\mathbb{R}^{2N}$ with a 2-form which is in every point constantly equal to Ω). It thus suffices to study the space $(\mathbb{R}^{2N}, \Omega)$ when dealing with symplectic integrators—for a particular physical problem, the symplectic manifold related with the Hamiltonian formulation of the problem can be locally identified with $(\mathbb{R}^{2N}, \Omega)$, so some symplectic properties will not be lost when working in the local coordinates instead of working directly on the symplectic manifold.

The next definition is motivated by the notion of symplectomorphism as a linear isomorphism preserving the symplectic structure.

Definition 10. Let $G \subset \mathbb{R}^{2N}$ be open and $\phi : G \to \mathbb{R}^{2N}$. We call ϕ a *symplectomorphism* (or *symplectic mapping/canonical transformation*), if ϕ is a diffeomorphism and for every $y \in G$, it holds that
$$(D\phi(y))^TJ(D\phi(y)) = J$$
($D\phi(y)$ denotes the Jacobian matrix of the mapping ϕ at the point y).

For a linear mapping ϕ represented by a matrix $S \in \mathbb{R}^{2N\times 2N}$, we have $D\phi(y) = S$ for any point $y \in \mathbb{R}^{2N}$. Thus, if ϕ is an isomorphism, then it is also a diffeomorphism and the property in the previous definition states $S^TJS = J$, consistently with the previous definition for linear symplectomorphisms.

Also note that the set of all symplectomorphisms on $\mathbb{R}^{2N}$ form a group under composition: diffeomorphisms form a group under composition; identity is clearly a symplectomorphism; for a symplectomorphism ϕ, its inverse ϕ^{-1} is a symplectomorphism, because from the inverse function theorem,

$$D(\phi^{-1}) = (D\phi \circ \phi^{-1})^{-1},$$

so the symplecticity of $D(\phi^{-1})$ in $y \in \mathbb{R}^{2N}$ follows from the symplecticity of the matrix $(D\phi)(\phi^{-1}(y))$, and thus also of its inverse, as symplectic matrices form a group. Finally, if ϕ and θ are symplectomorphisms, then from the chain rule, $D(\phi \circ \theta) = ((D\phi) \circ \theta)D\theta$ is symplectic as it is a product of two symplectic matrices.

Symplectic geometry in the Hamiltonian formalism

Let us now apply these concepts to the Hamiltonian formalism. First note that Hamilton's canonical equations (2.3) can now be written more compactly as

$$\dot{y} = J\nabla_y \mathcal{H}(t, y), \tag{2.4}$$

with the convention $y = (q^1, \dots q^n, p_1, \dots, p_n)$.

Much more important is this theorem (we follow the proof from [4], Chapter VI, Theorem 2.4):

Theorem 3. *Let $\mathcal{H}$ be a $\mathcal{C}^2$ Hamiltonian independent on time and defined on open $G \subset \mathbb{R}^{2N}$. Then the ODE flow φ_t given by the system (2.4) is symplectic.*

Proof. Let $[0, \tau]$ be a time interval such that for every $y_0 \in G$, the solution y_{y_0} to the IVP

$$\dot{y} = J\nabla\mathcal{H}(y), \qquad y(0) = y_0$$

exists, is unique and stays in G for all times from $[0, \tau]$. This is required to have a well defined flow $\varphi_t : G \to G$ for any $t \in [0, \tau]$.

Let $t \in (0, \tau)$ be fixed and let $y_0 \in G$ be arbitrary. It always holds that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(y_0) = J\nabla\mathcal{H}(\varphi_t(y_0))$$

as φ_t describes solutions to the IVPs. Let $\psi_t(y_0) := D\varphi_t(y_0)$, we want to show that $\psi_t^T J\psi_t = J$.

ψ_t satisfies the variational equation

$$\dot{\psi}_t(y_0) = J\Big(D\nabla\mathcal{H}(\varphi_t(y_0))\Big)\psi_t(y_0)$$

from the standard method of differentiating ODE solution with respect to initial conditions (which can be formally derived by applying D operator to $\dot{y} = J\nabla\mathcal{H}(y)$, interchanging derivatives on the left hand side of the equation, using the fact that the operator satisfies $DJ = JD$, as J is a constant matrix, and finally using the chain rule for $\mathcal{H} \circ \varphi_t$).

The expression $D\nabla\mathcal{H}(\varphi_t(y_0))$ is just the Hessian matrix $D^2\mathcal{H}(\varphi_t(y_0)) = (\partial_r\partial_s\varphi_t(y_0))_{r,s}$, which is symmetric, because $\mathcal{H} \in \mathcal{C}^2(G)$, which implies $\varphi_t \in \mathcal{C}^2(G)$, and thus the derivatives are interchangeable. Therefore,

$$\dot{\psi}_t(y_0) = J(D^2\mathcal{H}(\varphi_t(y_0)))\psi_t(y_0).$$

Now, denoting $y = \varphi_t(y_0)$, we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(D\varphi_t(y_0)^T J D\varphi_t(y_0)) &= \frac{\mathrm{d}}{\mathrm{d}t}(\psi_t^T(y_0) J \psi_t(y_0)) = \dot{\psi}_t^T(y_0) J \psi_t(y_0) + \psi_t^T(y_0) J \dot{\psi}_t(y_0) \\ &= (J D^2\mathcal{H}(y)\psi_t(y_0))^T J \psi_t(y_0) + \psi_t^T(y_0) J J D^2\mathcal{H}(y)\psi_t(y_0) \\ &= \psi_t^T(y_0) D^2\mathcal{H}(y) J^T J \psi_t(y_0) + \psi_t^T(y_0)(-I) D^2\mathcal{H}(y)\psi_t(y_0) \\ &= \psi_t^T(y_0) D^2\mathcal{H}(y)\psi_t(y_0) - \psi_t^T(y_0) D^2\mathcal{H}(y)\psi_t(y_0) = O,\end{aligned}$$

using the relations $J^2 = -I$, $J^T J = I$ and the symmetry of the Hessian matrix.

Therefore, for any $y_0 \in G$, the function $t \mapsto D\varphi_t(y_0)^T J D\varphi_t(y_0)$ is constant over $[0, \tau]$ (as it is also continuous). Because

$$D\varphi_0(y_0)^T J D\varphi_0(y_0) = I^T J I = J,$$

we finally have

$$D\varphi_t(y_0)^T J D\varphi_t(y_0) = J$$

for arbitrary $y_0 \in G$ and $t \in [0, \tau]$, so the flow is symplectic. $\square$

It can be even shown that this property characterises the Hamiltonian systems—if the flow of an ODE is symplectic, then it is (at least locally) a Hamiltonian system (see [4], Chapter VI, Theorem 2.6), which indicates a deep connection between the Hamiltonian systems and symplectic geometry.

Finally, we mention the important property that the symplecticity of the flow φ_t implies that the flow preserves volumes: For a measurable set $M \subset \mathbb{R}^{2N}$, the substitution theorem gives

$$\mathrm{vol}(\varphi_t(M)) = \int_{\varphi_t(M)} \mathrm{d}x = \int_M |\mathrm{Jac}\ \varphi_t(x)| dx = \int_M dx = \mathrm{vol}(M),$$

because the Jacobian of the flow satisfies $|\mathrm{Jac}\ \varphi_t(x)| = 1$, which follows from the symplecticity condition:

$$\det J = \det\Big((D\varphi_t)^T J (D\varphi_t)\Big) = \Big(\det(D\varphi_t)\Big)^2 \cdot \det J = (\mathrm{Jac}\ \varphi_t)^2 \cdot \det J$$

implies $|\mathrm{Jac}\ \varphi_t| = 1$ (because $\det J \neq 0$).

Alternatively, utilising the structure of the ODE (2.4) instead of its flow, the preservation of volumes follows from the fact that the vector field $J\nabla_z\mathcal{H}$ from the equation $\dot{y} = J\nabla_z\mathcal{H}(y)$ is divergence-free ([3], Theorem 14.8). This can be simply verified:

$$\mathrm{div}(J\nabla_z\mathcal{H}) = \nabla_{(q,p)} \cdot \begin{pmatrix} \nabla_p\mathcal{H} \\ -\nabla_q\mathcal{H} \end{pmatrix} = \sum_{i=1}^N \partial_{q_i}(\partial_{p_i}\mathcal{H}) + \sum_{i=1}^N \partial_{p_i}(-\partial_{q_i}\mathcal{H}) = 0.$$

3 Symplectic integrators

In the previous chapter, we showed that the Hamiltonian flow given by the autonomous system $\dot{y} = J\nabla\mathcal{H}(y)$ is symplectic. However, employing a particular discretisation, the numerical flow defined in Chapter 1 does not in general have to be symplectic. Yet, there exist numerical schemes whose corresponding numerical flows are symplectic for any Hamiltonian and we shall see that this property has several advantageous numerical consequences. Here, we follow mainly texts [4, 5, 6, 11].

Definition 11. Let $h > 0$ and let Ψ_h be a step function of a particular numerical scheme for solving Hamiltonian systems. For a smooth time-independent Hamiltonian $\mathcal{H}$, let $\Phi_h^{\mathcal{H}}$ denote the numerical flow corresponding to the scheme Ψ_h and applied to the system (2.4). The scheme given by Ψ_h is said to be *symplectic*, if for every smooth Hamiltonian $\mathcal{H}$, the numerical flow $\Phi_h^{\mathcal{H}}$ is a symplectomorphism.

Example. The *Symplectic Euler method* is given by the equations

$$\begin{aligned} p_{n+1} &= p_n - h\nabla_q\mathcal{H}(q_n, p_{n+1}), \\ q_{n+1} &= q_n + h\nabla_p\mathcal{H}(q_n, p_{n+1}). \end{aligned} \tag{3.1}$$

This can be viewed as using the Implicit Euler method for the momenta step and the Explicit Euler method for the positions step.

Let us check that the numerical flow

$$\Phi_h : (q_n, p_n) \mapsto (q_{n+1}, p_{n+1})$$

is symplectic for any smooth Hamiltonian $\mathcal{H}$. For brevity, we shall denote $q := q_n$, $p := p_n$ the old coordinates, $\tilde{q} := q_{n+1}$, $\tilde{p} := p_{n+1}$ the new coordinates, I and O the identity and zero matrices of size N, respectively, and we shall use subscripts for the derivatives, e.g. $\mathcal{H}_p := \nabla_p\mathcal{H}(q, \tilde{p})$, $\mathcal{H}_{qp} := \nabla_q\nabla_p\mathcal{H}(q, \tilde{p})$ or $\tilde{p}_q := \nabla_q\tilde{p}$ (we always evaluate $\mathcal{H}$ and its derivatives in $(q, \tilde{p})$). We presume $\mathcal{H}$ is smooth enough so that all the partial derivatives used in our calculations exist and are interchangeable. Note that $\mathcal{H}_{pp}$ and $\mathcal{H}_{qq}$ are symmetric $N \times N$ matrices and that

$$\mathcal{H}_{pq} = \left(\frac{\partial^2\mathcal{H}}{\partial p_s \partial q_r}\right)_{r,s} = \left(\frac{\partial^2\mathcal{H}}{\partial q_s \partial p_r}\right)_{r,s}^T = \mathcal{H}_{qp}^T.$$

In this notation, the scheme is given by

$$\tilde{p} = p - h\mathcal{H}_q, \qquad \tilde{q} = q + h\mathcal{H}_p.$$

Differentiating the equations defining the Symplectic Euler method, we have

$$\tilde{q}_p = h\mathcal{H}_{pp}\tilde{p}_p, \tag{3.2}$$

$$\tilde{q}_q = I + h\mathcal{H}_{qp} + h\mathcal{H}_{pp}\tilde{p}_q, \tag{3.3}$$

$$\tilde{p}_q = -h\mathcal{H}_{qq} - h\mathcal{H}_{pq}\tilde{p}_q, \tag{3.4}$$

$$\tilde{p}_p = I - h\mathcal{H}_{pq}\tilde{p}_p, \tag{3.5}$$

so from the third and fourth equation,

$$(I + h\mathcal{H}_{pq})\tilde{p}_q = -h\mathcal{H}_{qq}, \tag{3.6}$$
$$(I + h\mathcal{H}_{pq})\tilde{p}_p = I. \tag{3.7}$$

Let us now rewrite the expression from the symplecticity condition:

$$(D\Phi_h)^T J(D\Phi_h) = \begin{pmatrix} \tilde{q}_q^T & \tilde{p}_q^T \\ \tilde{q}_p^T & \tilde{p}_p^T \end{pmatrix} \begin{pmatrix} O & I \\ -I & O \end{pmatrix} \begin{pmatrix} \tilde{q}_q & \tilde{q}_p \\ \tilde{p}_q & \tilde{p}_p \end{pmatrix} = \begin{pmatrix} \tilde{q}_q^T & \tilde{p}_q^T \\ \tilde{q}_p^T & \tilde{p}_p^T \end{pmatrix} \begin{pmatrix} \tilde{p}_q & \tilde{p}_p \\ -\tilde{q}_q & -\tilde{q}_p \end{pmatrix},$$

where we view $(\tilde{q}, \tilde{p})$ as the actual functions defining the numerical flow Φ_h; this matrix product needs to yield the matrix J in order to show that the Symplectic Euler method is symplectic.

The first block of the resulting matrix product is

$$\begin{aligned} \tilde{q}_q^T\tilde{p}_q - \tilde{p}_q^T\tilde{q}_q &= (I + h\mathcal{H}_{qp} + h\mathcal{H}_{pp}\tilde{p}_q)^T\tilde{p}_q - \tilde{p}_q^T(I + h\mathcal{H}_{qp} + h\mathcal{H}_{pp}\tilde{p}_q) = \\ &= (I + h\mathcal{H}_{qp}^T)\tilde{p}_q + h\tilde{p}_q^T\mathcal{H}_{pp}^T\tilde{p}_q - ((I + h\mathcal{H}_{qp})^T\tilde{p}_q)^T - \tilde{p}_q^T h\mathcal{H}_{pp}\tilde{p}_q = \\ &= (I + h\mathcal{H}_{pq})\tilde{p}_q - ((I + h\mathcal{H}_{pq})\tilde{p}_q)^T = -h\mathcal{H}_{qq} + h\mathcal{H}_{qq}^T = O, \end{aligned}$$

where we used (3.3) and (3.6).

The block in the first row and the second column is

$$\begin{aligned} \tilde{q}_q^T\tilde{p}_p - \tilde{p}_q^T\tilde{q}_p = (I + h\mathcal{H}_{qp} + h\mathcal{H}_{pp}\tilde{p}_q)^T\tilde{p}_p - \tilde{p}_q^T h\mathcal{H}_{pp}\tilde{p}_p = (I + h\mathcal{H}_{pq})\tilde{p}_p + \\ + h\tilde{p}_q^T\mathcal{H}_{pp}^T\tilde{p}_p - \tilde{p}_q^T h\mathcal{H}_{pp}\tilde{p}_p = I + O = I, \end{aligned}$$

where we used (3.3), (3.2) and (3.7).

The block in the second row and the first column is

$$\tilde{q}_p^T\tilde{p}_q - \tilde{p}_p^T\tilde{q}_q = -(\tilde{q}_q^T\tilde{p}_p - \tilde{p}_q^T\tilde{q}_p)^T = -I^T = -I$$

thanks to the previous computation.

Finally, the block in the second row and second column is

$$\tilde{q}_p^T\tilde{p}_p - \tilde{p}_p^T\tilde{q}_p = h\tilde{p}_p^T\mathcal{H}_{pp}^T\tilde{p}_p - \tilde{p}_p^T h\mathcal{H}_{pp}\tilde{p}_p = O$$

from (3.2).

Therefore, the Symplectic Euler method is symplectic, because it holds that $(D\Phi_h)^T J(D\Phi_h) = J$ for arbitrary smooth Hamiltonian.

The related scheme

$$\begin{aligned} q_{n+1} &= q_n + h\nabla_p\mathcal{H}(q_{n+1}, p_n), \\ p_{n+1} &= p_n - h\nabla_q\mathcal{H}(q_{n+1}, p_n). \end{aligned} \tag{3.8}$$

is also called Symplectic Euler method. The symplecticity can be either checked analogously as in the previous example, or one can reason that this scheme (3.8) is equivalent to using the scheme (3.1) "backwards" with step size $-h$ (i.e. $\Phi_{-h}^{-1} : (q_{n+1}, p_{n+1}) \mapsto (q_n, p_n)$ instead of $\Phi_h : (q_n, p_n) \mapsto (q_{n+1}, p_{n+1})$). Because symplectomorphisms form a group, the symplecticity of Φ_{-h}^{-1}, and thus of scheme (3.8) follows.

Another symplectic integrator is the *Störmer-Verlet scheme*

$$\begin{aligned} p_{n+1/2} &= p_n - \frac{h}{2}\nabla_q\mathcal{H}(q_n, p_{n+1/2}), \\ q_{n+1} &= q_n + \frac{h}{2}\Big(\nabla_p\mathcal{H}(q_n, p_{n+1/2}) + \nabla_p\mathcal{H}(q_{n+1}, p_{n+1/2})\Big), \\ p_{n+1} &= p_{n+1/2} - \frac{h}{2}\nabla_q\mathcal{H}(q_{n+1}, p_{n+1/2}). \end{aligned} \tag{3.9}$$

Again, its symplecticity could be shown similarly as in the case of Symplectic Euler method, but there is a shorter way—at first, we perform one step of the scheme (3.1) with step size $h/2$, moving from the point (q_n, p_n) to the point $(q_{n+1/2}, p_{n+1/2})$; then, we use the scheme (3.8) to move from $(q_{n+1/2}, p_{n+1/2})$ to (q_{n+1}, p_{n+1}). This gives us the relations

$$\begin{aligned} p_{n+1/2} &= p_n - \frac{h}{2}\nabla_q\mathcal{H}(q_n, p_{n+1/2}), \\ q_{n+1/2} &= q_n + \frac{h}{2}\nabla_p\mathcal{H}(q_n, p_{n+1/2}), \\ q_{n+1} &= q_{n+1/2} + \frac{h}{2}\nabla_p\mathcal{H}(q_{n+1}, p_{n+1/2}), \\ p_{n+1} &= p_{n+1/2} - \frac{h}{2}\nabla_q\mathcal{H}(q_{n+1}, p_{n+1/2}). \end{aligned}$$

Now, the first and the last equations correspond exactly to the first and last equations in (3.9), and plugging the second equation into the third equation yields the second equation in (3.9). Therefore, Störmer-Verlet method is actually a composition of the two previously presented Symplectic Euler methods. Because symplectomorphisms form a group under composition, the Störmer-Verlet method is symplectic.

Symplectic methods of arbitrary high order can be defined. A common method for that is to compose several Symplectic Euler methods steps with different step sizes so that the resulting composition has a larger order, while preserving the symplecticity (just as in the case of the Störmer-Verlet method).

For Runge-Kutta methods (1.4), it can be shown that a sufficient condition on their symplecticity is

$$a_{ij}b_i + a_{ji}b_j = b_ib_j, \quad \forall i, j \in \{1, \dots, s\},$$

but their complete characterisation is more complicated (see Chapter VI, Theorem 4.3 in [4] for the sufficient condition and Section VI.7 for the characterisation). More commonly, symplectic integrators are partitioned Runge-Kutta methods, as they naturally reflect the difference between positions and momenta coordinates. In this setting, the Symplectic Euler scheme (3.1) has the form

$$\begin{aligned} p_{n+1} &= p_n - h\nabla_q\mathcal{H}(q_n, p_{n+1}) = p_n + h\hat{b}_1\ell_1, \\ q_{n+1} &= q_n + h\nabla_p\mathcal{H}(q_n, p_{n+1}) = q_n + hb_1k_1, \\ \ell_1 &= -\nabla_q\mathcal{H}(q_n, p_{n+1}) = -\nabla_q\mathcal{H}(q_n, p_n + h\ell_1), \\ k_1 &= \nabla_p\mathcal{H}(q_n, p_{n+1}) = \nabla_p\mathcal{H}(q_n, p_n + h\ell_1), \text{ so} \\ b_1 &= 1, \quad \hat{b}_1 = 1, \quad a_{11} = 0, \quad \hat{a}_{11} = 1. \end{aligned}$$

Both the dual Symplectic Euler scheme (3.8) and the Störmer-Verlet scheme (3.9) are also partitioned Runge-Kutta methods.

All of the presented schemes (3.1), (3.8) and (3.9) are implicit, even though they are among the simplest symplectic schemes. This is actually the case for most of the symplectic integrators, at least if a general Hamiltonian $\mathcal{H}$ is considered.

If the Hamiltonian is assumed to have a particular structure, explicit schemes can be designed. This is the case for a very common class of separable Hamiltonians, which we have already introduced in Definition 6. Hamiltonians describing many mechanical systems are of this form, at least when described in Cartesian[1] coordinates. In those cases,

$$\mathcal{H}(q,p) = Q(q) + P(p), \qquad \nabla_q \mathcal{H}(q,p) = \nabla_q Q(q), \qquad \nabla_p \mathcal{H}(q,p) = \nabla_p P(p),$$

which for example means that the momentum update in (3.1) is actually explicit:

$$p_{n+1} = p_n - h\nabla_q \mathcal{H}(q_n, p_{n+1}) = p_n - h\nabla_q Q(q_n).$$

Similarly, the other two introduced schemes (3.8) and (3.9) are explicit in the case of separable Hamiltonian.

For more complicated (partitioned) Runge-Kutta methods, the separability of the Hamiltonian, however, does not guarantee that the scheme will be explicit. There are also many physical systems whose Hamiltonians are not separable, e.g. systems with electromagnetic fields, which we study in Chapter 5. Therefore, the general implicitness of the symplectic schemes needs to be considered and we shall do it in Chapter 4.

3.1 Backward analysis and approximate conservation of energy

In this section, we will utilise the backward error analysis from Section 1.2.2 to show that the Hamiltonian is surprisingly well preserved when the employed numerical scheme is symplectic. This is not obvious, as the definition of symplectic integrators does not explicitly aim to preserve anything else than the matrix of the symplectic structure J. On the other hand, we already commented that the symplecticity of an ODE flow characterises Hamiltonian systems and that is the underlying idea behind the preservation of the Hamiltonian.

We start with the following lemma, see [4], Chapter VI, Lemma 2.7:

Lemma 4. *Let $G \subset \mathbb{R}^N$ be open, $f \in \mathcal{C}^1(G, \mathbb{R}^N)$ and for all $y \in G$, $Df(y)$ is a symmetric matrix. Then, for every $a \in G$ there exists a neighbourhood $U(a) \subset G$ of a and $\mathcal{H} \in \mathcal{C}^2(U(a), \mathbb{R})$ such that $f = \nabla \mathcal{H}$ on $U(a)$.*

Proof. Let $a \in G$ and $U(a)$ be an open ball so that $U(a) \subset G$. Let

$$\mathcal{H}(y) := \int_0^1 (y-a)^T f(a + t(y-a))\mathrm{d}t, \quad y \in U(a).$$

[1] The usual separation has the form $\mathcal{H}(q,p) = T(p) + V(q)$, where T is the kinetic and V is the potential energy of the system. In curvilinear coordinates, however, the kinetic energy can also depend on positions (e.g. for a single particle with mass m described in polar coordinates (r,θ), its kinetic energy is $T = \frac{1}{2m}(p_r^2 + (\frac{p_\theta}{r})^2)$.

This function is well defined: $U(a)$ contains the whole line segment

$$\{a + t(y - a) \mid t \in [0, 1]\}$$

with endpoints $a, y \in U(a)$, this line segment is compact and

$$t \mapsto (y - a)^T f(a + t(y - a))$$

is continuous there, so it is an integrable function. Moreover, for $k \in \{1, \dots, N\}$ arbitrary fixed and $y \in U(a)$, there is a closed ball $B(y) \subset U(a)$ and

$$\partial_k((z - a)^T f(a + t(z - a))$$

is in particular continuous on the compact $[0, 1] \times B(y)$ with respect to both t and z, so for any $t \in [0, 1]$ and $z \in B(y)$, this derivative can be estimated by the same constant, which is of course integrable on $[0, 1]$. From Lebesgue's theorem, it follows that we can interchange the integral and partial derivative as

$$\begin{aligned} \partial_k \mathcal{H}(y) &= \int_0^1 \partial_k \Big((y - a)^T f(a + t(y - a))\Big) \mathrm{d}t = \\ &= \int_0^1 \Big(f_k(a + t(y - a)) + (y - a)^T Df(a + t(y - a)) t e_k\Big) \mathrm{d}t, \end{aligned}$$

where e_k is the k-th canonical basis vector.

We have

$$Df(a + t(y - a))e_k = \partial_k f(a + t(y - a)) = \nabla f_k(a + t(y - a))$$

from the assumption that $Df(a + t(y - a))$ is symmetric. Moreover,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(t f_k(a + t(y - a))) &= f_k(a + t(y - a)) + t \sum_{i=1}^{N} \partial_i f_k(a + t(y - a))(y_i - a_i) = \\ &= f_k(a + t(y - a)) + t(y - a)^T \nabla f_k(a + t(y - a)). \end{aligned}$$

Therefore,

$$\begin{aligned} \partial_k \mathcal{H}(y) &= \int_0^1 \Big(f_k(a + t(y - a)) + t(y - a)^T \nabla f_k(a + t(y - a))\Big) \mathrm{d}t = \\ &= \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\Big(t f_k(a + t(y - a))\Big) \mathrm{d}t = [t f_k(a + t(y - a))]_0^1 = f_k(y). \end{aligned}$$

Because this holds for every k, we have $f = \nabla \mathcal{H}$ on $U(a)$. $\square$

From the proof, we also see that if $G = \mathbb{R}^N$, there exists $\mathcal{H}$ satisfying $f = \nabla \mathcal{H}$ globally on the whole $\mathbb{R}^N$.

We are now ready to prove the next theorem, giving us a connection between symplectic numerical methods and modified equations. We follow the proof from [4], Chapter IX, Theorem 3.1:

Theorem 5 (Modified equation of symplectic scheme)**.** *Let $K \in \mathbb{N}$, let $\mathcal{H}$ be a smooth time-independent Hamiltonian on $\mathbb{R}^{2N}$ and consider a consistent symplectic numerical scheme with numerical flow Φ_h and K-truncated modified equation of the form* (1.7) *with smooth f_i, $i \in \{0, \dots, K\}$. Then, for $k \in \{0, \dots, K\}$, there exists smooth $\mathcal{H}_k$ defined on $\mathbb{R}^{2N}$ such that $f_k = J\nabla \mathcal{H}_k$ on $\mathbb{R}^{2N}$.*

Proof. We will prove this by induction. The base case for $k = 0$ stems from the fact that the numerical scheme is assumed to be consistent, so $f_0 = J\nabla\mathcal{H}$ (the right hand side of the original ODE (2.4)), and thus $\mathcal{H}_0 = \mathcal{H}$.

For the induction step, assume that $f_j = J\nabla\mathcal{H}_j$ for $j \in \{0, \dots, k\}$. Let

$$\mathcal{H}^{[k]} := \mathcal{H}_0 + h\mathcal{H}_1 + \cdots + h^k\mathcal{H}_k.$$

Then, the k-truncated modified equation has the form

$$y' = J\nabla\mathcal{H}_0 + hJ\nabla\mathcal{H}_1 + \cdots + h^k J\nabla\mathcal{H}_k = J\nabla\mathcal{H}^{[k]},$$

so it is a Hamiltonian system and its flow $\varphi_h^{[k]}$ is symplectic.

Let $y \in \mathbb{R}^{2N}$. From (1.8), we have

$$\Phi_h(y) = \varphi_h^{[k]}(y) + h^{k+1}f_{k+1}(y) + \mathcal{O}(h^{k+2}), \quad h \to 0.$$

Therefore also

$$D\Phi_h(y) = D\varphi_h^{[k]}(y) + h^{k+1}Df_{k+1}(y) + \mathcal{O}(h^{k+2}), \quad h \to 0.$$

Moreover, $D\varphi_h^{[k]}(y) = I + \mathcal{O}(h)$, $h \to 0$: that is because

$$\varphi_h^{[k]}(y) = \tilde{y}(h) = y + h\tilde{y}'(\xi_h) = y + hJ\nabla\mathcal{H}^{[k]}(\tilde{y}(\xi_h))$$

for a $\xi_h \in [0, h]$ from Lagrange remainder theorem ($\tilde{y}$ is the solution of the k-truncated modified equation $y' = J\nabla\mathcal{H}^{[k]}$ with $\tilde{y}(0) = y$), and therefore also

$$D\varphi_h^{[k]}(y) = I + hDJ\nabla\mathcal{H}^{[k]}(\tilde{y}(\xi_h)) = I + hJD\nabla\mathcal{H}^{[k]}(\tilde{y}(\xi_h)) = I + \mathcal{O}(h), \quad h \to 0,$$

because $\xi_h \to 0$ as $h \to 0$, $D\nabla\mathcal{H}^{[k]}$ is continuous ($\mathcal{H}^{[k]}$ is smooth from the induction hypothesis) and $\tilde{y}$ is also continuous as a solution of an ODE.

Because $\varphi^{[k]}$ and Φ_h are both symplectic, we thus have (all functions are evaluated in y)

$$\begin{aligned}
J &= (D\Phi_h)^T J(D\Phi_h) = \\
&= (D\varphi_h^{[k]} + h^{k+1}Df_{k+1} + \mathcal{O}(h^{k+2})^T J(D\varphi_h^{[k]} + h^{k+1}Df_{k+1} + \mathcal{O}(h^{k+2})) = \\
&= (D\varphi_h^{[k]})^T J(D\varphi_h^{[k]}) + h^{k+1}((Df_{k+1})^T J(D\varphi_h^{[k]}) + (D\varphi_h^{[k]})^T J(Df_{k+1})) + \mathcal{O}(h^{k+2})
\end{aligned}$$

and because $(D\varphi_h^{[k]})^T J(D\varphi_h^{[k]}) = J$ and by comparing the left hand side and the right hand side of the previous equation, which holds for all $h > 0$, we have that

$$(Df_{k+1})^T J(D\varphi_h^{[k]}) + (D\varphi_h^{[k]})^T J(Df_{k+1}) = O,$$

O being the zero matrix.

Then, from $D\varphi_h^{[k]} = I + \mathcal{O}(h)$, the previous expression can be written as

$$(Df_{k+1})^T J + J(Df_{k+1}) + \mathcal{O}(h) = O, \quad h \to 0,$$

so it also has to hold (h is arbitrary)

$$(Df_{k+1})^T J + J(Df_{k+1}) = O,$$

which can be rewritten as

$$D(Jf_{k+1}) = -(Df_{k+1})^T J = (Df_{k+1})^T J^T = (D(Jf_{k+1}))^T,$$

because $J^T = -J$. Therefore, $D(Jf_{k+1})(y)$ is symmetric for arbitrary y. From the previous Lemma 4, there exists $-\mathcal{H}_{k+1}$ such that $Jf_{k+1} = \nabla(-\mathcal{H}_{k+1})$, which completes the induction step, as

$$f_{k+1} = J^{-1}\nabla(-\mathcal{H}_{k+1}) = -J^{-1}\nabla\mathcal{H}_{k+1} = J\nabla\mathcal{H}_{k+1}.$$

□

The previous theorem is interesting on its own, as it shows that for symplectic numerical schemes, the symplectic properties are preserved when moving from the original ODE to the modified ODE. It is also the key for proving the next two theorems, which describe the almost-preservation of energy by symplectic schemes. The first theorem shows how the energy is perturbed for the modified Hamiltonian due to the truncation of the modified equation. This result is then used in the proof of the second theorem, which estimates the energy perturbation of the original system with the help of the truncated modified equation. In both proofs, we follow [4], Chapter IX, Theorem 8.1, but we use a simpler estimate on $|\Phi_h - \varphi_h^{[K]}|$ than the one presented in the book. With stronger assumptions (analytic $\mathcal{H}$), better results can be obtained.

Theorem 6 (Near-conservation of the modified Hamiltonian)**.** *Let $G \subset \mathbb{R}^{2N}$ be open, $\mathcal{H} : G \to \mathbb{R}$ be a smooth time-independent Hamiltonian, Φ_h be a numerical flow of a symplectic method giving approximations $(y_i)_{i=0}^n$ and assume that its truncated modified equation admits a smooth modified Hamiltonian $\mathcal{H}^{[K]}$ defined on G. Let $C \subset G$ be compact and $y_i \in C$, $i \in \{0, \dots, n\}$. Then*

$$\mathcal{H}^{[K]}(y_n) = \mathcal{H}^{[K]}(y_0) + \mathcal{O}(nh^{K+1}). \tag{3.10}$$

Proof. Let $\varphi_t^{[K]}$ denote the numerical flow of the truncated modified equation with the form $y' = J\nabla\mathcal{H}^{[K]}$ thanks to Theorem 5. From (1.8), it holds that

$$\|y_{i+1} - \varphi_h^{[K]}(y_i)\| = \|\Phi_h(y_i) - \varphi_h^{[K]}(y_i)\| = \mathcal{O}(h^{K+1}), \quad h \to 0,$$

for $i \in \{0, \dots, n-1\}$. From the differentiability of $\mathcal{H}^{[K]}$ on G and compactness of $C \subset G$, $\mathcal{H}^{[K]}$ is Lipschitz continuous on C, therefore also

$$|\mathcal{H}^{[K]}(y_{i+1}) - \mathcal{H}^{[K]}(\varphi_h^{[K]}(y_i))| = \mathcal{O}(1) \cdot \|y_{i+1} - \varphi_h^{[K]}(y_i)\| = \mathcal{O}(h^{K+1}).$$

Because $\varphi_h^{[K]}$ is the exact flow of equation $y' = J\nabla\mathcal{H}^{[K]}$ with time-independent Hamiltonian, it follows from Theorem 2 that

$$\mathcal{H}^{[K]}(\varphi_h^{[K]}(y_i)) = \mathcal{H}^{[K]}(y_i).$$

Therefore,

$$\begin{aligned}
\mathcal{H}^{[K]}(y_n) - \mathcal{H}^{[K]}(y_0) &= \sum_{i=0}^{n-1}(\mathcal{H}^{[K]}(y_{i+1}) - \mathcal{H}^{[K]}(y_i)) = \\
&= \sum_{i=0}^{n-1}(\mathcal{H}^{[K]}(y_{i+1}) - \mathcal{H}^{[K]}(\varphi_h^{[K]}(y_i))) = \\
&= \sum_{i=0}^{n-1}\mathcal{O}(h^{K+1}) = \mathcal{O}(nh^{K+1}), \quad h \to 0.
\end{aligned}$$

□

Theorem 7 (Near-conservation of the original Hamiltonian)**.** *Let $G \subset \mathbb{R}^{2N}$ be open, $\mathcal{H} : G \to \mathbb{R}$ be a smooth time-independent Hamiltonian, Φ_h be a numerical flow of a p-order symplectic method giving approximations $(y_i)_{i=0}^n$, $p \leq K$ and assume that its K-truncated modified equation admits a smooth modified Hamiltonian $\mathcal{H}^{[K]}$ defined on G. Let $C \subset G$ be compact and $y_i \in C$, $i \in \{0, \dots, n\}$. Then*

$$\mathcal{H}(y_n) = \mathcal{H}(y_0) + \mathcal{O}(h^p) + \mathcal{O}(nh^{K+1}), \quad h \to 0. \tag{3.11}$$

Proof. Using the triangle inequality,

$$|\mathcal{H}(y_n) - \mathcal{H}(y_0)| \leq |\mathcal{H}(y_n) - \mathcal{H}^{[K]}(y_n)| + |\mathcal{H}^{[K]}(y_n) - \mathcal{H}^{[K]}(y_0)| + |\mathcal{H}^{[K]}(y_0) - \mathcal{H}(y_0)|.$$

From the previous theorem, the second term is of order $\mathcal{O}(nh^{K+1})$, so it suffices to show that the first and the third terms are both of order $\mathcal{O}(h^p)$. Because the numerical scheme is symplectic and of order p, its truncated modified equation has the form (see Theorem 1)

$$y' = J\nabla\mathcal{H}^{[K]} = J\nabla(\mathcal{H} + h^p\mathcal{H}_p + h^{p+1}\mathcal{H}_{p+1} + \cdots + h^K\mathcal{H}_K),$$

i.e. $\mathcal{H}^{[K]} - \mathcal{H} = h^p\mathcal{H}_p + \cdots + h^K\mathcal{H}_K$. Therefore, for arbitrary $y \in C$, we have

$$|\mathcal{H}(y) - \mathcal{H}^{[K]}(y)| \leq h^p \left\| \mathcal{H}_p + \cdots + h^{K-p}\mathcal{H}_K \right\|_\infty,$$

where $\| \cdot \|_\infty$ is the supremum norm over C for y and some interval $[0, h_{\max}]$ for h (the norm is finite for the function

$$(y, h) \mapsto \mathcal{H}_p(y) + \cdots + h^{K-p}\mathcal{H}_K(y),$$

as it is continuous and $C \times [0, h_{\max}]$ is compact). Therefore,

$$|\mathcal{H}(y) - \mathcal{H}^{[K]}(y)| = h^p \cdot \mathcal{O}(1) = \mathcal{O}(h^p), \quad h \to 0.$$

In particular, both $|\mathcal{H}(y_n) - \mathcal{H}^{[K]}(y_n)|$ and $|\mathcal{H}(y_0) - \mathcal{H}^{[K]}(y_0)|$ are of order $\mathcal{O}(h^p)$. □

The previous theorem describes the almost-conservation of the original Hamiltonian. The term nh gives the total integration time t, so (3.11) can be also rewritten as

$$\mathcal{H}(y_n) = \mathcal{H}(y_0) + \mathcal{O}(h^p) + \mathcal{O}(th^K), \quad h \to 0. \tag{3.12}$$

This equation tells us that there are two error terms in the energy discrepancy, the first one being time-independent, whereas the second one growing linearly with integration time t. If we set the truncation index K larger than the order p, the first time-independent error term $\mathcal{O}(h^p)$ can be assumed to be dominant, compared to the error term $\mathcal{O}(th^K)$, at least for times $t \propto h^{p-K}$. We can force $h^{p-K} \gg 1$ by choosing suitable $h \in (0, 1)$ and $K > p$, thus having the energy almost-conserved with no linear drift on a very long time interval.

On the contrary, for a general non-symplectic numerical scheme, we only get the estimate

$$\mathcal{H}(y_n) = \mathcal{H}(y_0) + \mathcal{O}(th^p),$$

so the leading error term is already time-dependent with linear drift of the energy, even on short time intervals.

To show this, denote φ_h the exact ODE flow of the original Hamiltonian system. Then we know that

$$|\mathcal{H}(y_{i+1}) - \mathcal{H}(\varphi_h(y_i))| = \mathcal{O}(1)\|y_{i+1} - \varphi_h(y_i)\| = \mathcal{O}(h^{p+1})$$

from the same reasoning about Lipschitz continuity on compact sets as in the proof of Theorem 6 and from the definition of order. Because φ_h preserves $\mathcal{H}$, this gives

$$|\mathcal{H}(y_{i+1}) - \mathcal{H}(y_i)| = |\mathcal{H}(y_{i+1}) - \mathcal{H}(\varphi_h(y_i))| = \mathcal{O}(h^{p+1}),$$

and therefore again from a telescopic sum,

$$|\mathcal{H}(y_n) - \mathcal{H}(y_0)| \leq \sum_{i=0}^{n-1} |\mathcal{H}(y_{i+1}) - \mathcal{H}(y_i)| = \mathcal{O}(nh^{p+1}) = \mathcal{O}(th^p), \quad h \to 0.$$

4 Implicit integrators

For a general Hamiltonian, symplectic numerical schemes are usually implicit, which means that in every step of the numerical integration, a system of algebraic equations of the form (1.2) has to be solved. In some simple scenarios, such an equation can be analytically solved for a particular Hamiltonian and numerical scheme, but in general, a numerical root-finding algorithm has to be employed.

In this section, we shall study how the usage of one such algorithm affects the symplectic structure of the Symplectic Euler method, as it is the simplest case for analysis, yet it already demonstrates some problems which we assume to arise for any implicit symplectic integrators. These problems are usually neglected in literature, the presented results in this chapter are, to the best of our knowledge, novel.

We shall use the *fixed point iteration* (*FPI*) *method* for solving the systems of equations of the form $y = f(y)$, where $f : \mathbb{R}^N \to \mathbb{R}^N$. Other methods, such as Newton's method, could be also used.

The FPI method is very simple, as it chooses an initial approximation $y_0 \in \mathbb{R}^N$, and then it progressively computes $y_{n+1} = f(y_n)$. If f is a *contraction mapping* (i.e. it is L-Lipschitz continuous with $0 < L < 1$), then Banach's fixed point theorem guarantees that the previously defined sequence $(y_n)_{n=1}^{\infty}$ converges to a unique fixed point $y^* \in \mathbb{R}^N$ such that $y^* = f(y^*)$.

This method is a natural choice, because the system of algebraic equations we want to solve is already in the correct form $y_{n+1} = \Psi_h(y_n, y_{n+1}, t_n, t_{n+1})$ (where we now view y_n, t_n and t_{n+1} as known parameters) and because for many numerical schemes, $\Psi_h(y_n, \cdot, t_n, t_{n+1})$ is a contraction for step size h small enough.

In the following text, we shall study the Symplectic Euler method (3.1) combined with the FPI method. The complete integration method, going from the step (q, p) to the step $(\tilde{q}, \tilde{p})$, looks like this:

$$\begin{aligned} p_0 &:= p, \\ p_{j+1} &:= p - h\nabla_q \mathcal{H}(q, p_j), \quad j \in \{0, \dots, M-1\}, \\ \tilde{p} &:= p_M, \\ \tilde{q} &:= q + h\nabla_p \mathcal{H}(q, \tilde{p}). \end{aligned} \tag{4.1}$$

The method (4.1) uses M steps of the FPI method to find the new momentum (which is defined implicitly in the Symplectic Euler scheme (3.1)), and then it calculates the new position, as this update is defined explicitly in the scheme (3.1). This defines a new flow

$$\Phi_h^{[M]} : (q, p)^T \mapsto (\tilde{q}, p_M).$$

In practical implementation, a better stopping criterion should be used than a fixed number of fixed point iterations M. This is discussed in [12] together with the influence of computer finite precision arithmetic (FPA). We just note here that even though the FPI method perturbs the symplectic condition (as we shall see in the following sections), its effect can be minimised with enough steps M, because when the error between p_M and the true $\tilde{p}_{\text{true}}$ is of the order of machine epsilon, $\Phi_h^{[M]}$ is as close as it can be to the true Φ_h represented in FPA. Still, as [12] shows, the effect of FPA perturbs the symplecticity of Φ_h, which results in worse energy conservation properties, something we shall also see in Chapter 5.

4.1 Symplectic Euler's method in the plane

We shall now show our result on the order of the error in symplectic structure after one step of the scheme (4.1) in the case $N = 1$. Let us again use a short notation for the Hamiltonian and its derivatives as $\mathcal{H}^{(k)} := \mathcal{H}(q, p_k)$, $\mathcal{H}^{(M)}_{qp} := \partial_q \partial_p \mathcal{H}(q, p_M)$ (which is equal to $\mathcal{H}^{(M)}_{pq}$) and so on. We assume that the Hamiltonian is smooth-enough on $\mathbb{R}^2$, so that all the used derivatives are continuous, and also that the Hamiltonian and each of its relevant derivatives can be uniformly bounded, e.g. by restricting ourselves to just a compact and convex subset (for Taylor's expansions) of initial conditions (q, p), on which these uniform bounds will be obtained from continuity.

For arbitrary 2×2 matrix A, it holds that

$$A^T J A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & ad - bc \\ bc - ad & 0 \end{pmatrix} = \begin{pmatrix} 0 & \det A \\ -\det A & 0 \end{pmatrix},$$

therefore we are mainly interested in estimating the Jacobian of $\Phi_h^{[M]}$, i.e. the value

$$\iota_M := \partial_q \tilde{q} \cdot \partial_p \tilde{p} - \partial_q \tilde{p} \cdot \partial_p \tilde{q}.$$

From the explicit position update in (4.1), we have

$$\begin{aligned} \partial_q \tilde{q} &= \partial_q (q + h\mathcal{H}^{(M)}_p) = 1 + h(\mathcal{H}^{(M)}_{pq} \cdot 1 + \mathcal{H}^{(M)}_{pp} \partial_q p_M), \\ \partial_p \tilde{q} &= \partial_p (q + h\mathcal{H}^{(M)}_p) = h\mathcal{H}^{(M)}_{pp} \partial_p p_M, \end{aligned}$$

from which the Jacobian ι_M can be simplified as

$$\iota_M = (1 + h\mathcal{H}^{(M)}_{pq} + h\mathcal{H}^{(M)}_{pp} \partial_q \tilde{p}) \partial_p \tilde{p} - h \partial_q \tilde{p} \cdot \mathcal{H}^{(M)}_{pp} \partial_p \tilde{p} = (1 + h\mathcal{H}^{(M)}_{pq}) \partial_p p_M.$$

We shall show that $\iota_M = 1 + \mathcal{O}(h^{M+1})$, $h \to 0$.

Lemma 8. *It holds that*

$$\partial_p p_n = 1 - h\mathcal{H}^{(n-1)}_{pq} + h^2 \mathcal{H}^{(n-1)}_{pq} \mathcal{H}^{(n-2)}_{pq} - \cdots + (-1)^n h^n \mathcal{H}^{(n-1)}_{pq} \cdots \mathcal{H}^{(0)}_{pq},$$

or more compactly,

$$\partial_p p_n = \sum_{i=0}^{n} (-h)^i \prod_{j=1}^{i} \mathcal{H}^{(n-j)}_{pq}$$

(where the empty product $\prod_{j=1}^{0} a_j$ is defined as 1).

Proof. We will prove this by mathematical induction. For $n = 0$, the lemma says $\partial_p p_0 = 1$, which holds, because $p_0 = p$. The induction step follows from the recurrence for p_{n+1} in (4.1):

$$\begin{aligned} \partial_p p_{n+1} = \partial_p (p - h\mathcal{H}^{(n)}_q) = 1 - h\mathcal{H}^{(n)}_{pq} \partial_p p_n = 1 - h \sum_{i=0}^{n} (-h)^i \mathcal{H}^{(n)}_{pq} \prod_{j=1}^{i} \mathcal{H}^{(n-j)}_{pq} = \\ = \sum_{i=0}^{n+1} (-h)^i \prod_{j=1}^{i} \mathcal{H}^{(n+1-j)}_{pq}. \end{aligned}$$

$\square$

Lemma 9. *It holds that $p_n - p_{n-1} = \mathcal{O}(h^n)$, $h \to 0$.*

Proof. We will again use mathematical induction. For $n = 1$, direct substitution gives

$$p_1 - p_0 = (p - h\mathcal{H}_q^{(0)}) - p = -h\mathcal{H}_q^{(0)} = \mathcal{O}(h), \quad h \to 0$$

(this is why we assume that the derivatives of $\mathcal{H}$ have uniform bounds). In the induction step, we have

$$\begin{aligned} p_n - p_{n-1} &= (p - h\mathcal{H}_q^{(n-1)}) - (p - h\mathcal{H}_q^{(n-2)}) = -h(\mathcal{H}_q(q, p_{n-1}) - \mathcal{H}_q(q, p_{n-2})) = \\ &= -h(\mathcal{H}_q(q, p_{n-2}) + \mathcal{O}(p_{n-1} - p_{n-2}) - \mathcal{H}_q(q, p_{n-2})) = \\ &= -h\mathcal{O}(p_{n-1} - p_{n-2}) = -h \cdot \mathcal{O}(h^{n-1}) = \mathcal{O}(h^n), \qquad h \to 0, \end{aligned}$$

where we expanded $\mathcal{H}_q(q, p_{n-1})$ into Taylor's polynomial and used the induction hypothesis. □

Theorem 10. *After one step of the modified Symplectic Euler scheme in $\mathbb{R}^2$ given by (4.1) with M steps of the FPI method, the matrix of symplectic structure is perturbed as*

$$J_{new} = (1 + \mathcal{O}(h^{M+1}))J, \qquad \textit{where } J_{new} := \left(D\Phi_h^{[M]}\right)^T J \left(D\Phi_h^{[M]}\right).$$

Proof. It suffices to estimate ι_M:

$$\begin{aligned} \iota_M &= (1 + h\mathcal{H}_{pq}^{(M)})\partial_p p_M = (1 + h\mathcal{H}_{pq}^{(M)}) \sum_{i=0}^{M} (-h)^i \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} = \\ &= 1 + \sum_{i=1}^{M} (-h)^i \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} + \\ &\quad + h\mathcal{H}_{pq}^{(M)} \sum_{i=0}^{M-1} (-h)^i \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} + h\mathcal{H}_{pq}^{(M)} (-h)^M \prod_{j=1}^{M} \mathcal{H}_{pq}^{(M-j)} = \\ &= 1 - (-h)^{M+1} \prod_{j=0}^{M} \mathcal{H}_{pq}^{(j)} + \\ &\quad + \sum_{i=1}^{M} (-h)^i \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} - \sum_{i=0}^{M-1} (-h)^{i+1} \mathcal{H}_{pq}^{(M)} \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} \end{aligned} \tag{4.2}$$

and the last two sums can be rearranged as

$$\begin{aligned} \Delta_M &:= \sum_{i=1}^{M} (-h)^i \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} - \sum_{i=0}^{M-1} (-h)^{i+1} \mathcal{H}_{pq}^{(M)} \prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} = \\ &= \sum_{i=1}^{M} (-h)^i \left(\prod_{j=1}^{i} \mathcal{H}_{pq}^{(M-j)} - \mathcal{H}_{pq}^{(M)} \prod_{j=1}^{i-1} \mathcal{H}_{pq}^{(M-j)} \right) = \\ &= \sum_{i=1}^{M} (-h)^i (\mathcal{H}_{pq}^{(M-i)} - \mathcal{H}_{pq}^{(M)}) \prod_{j=1}^{i-1} \mathcal{H}_{pq}^{(M-j)}. \end{aligned} \tag{4.3}$$

Using a telescopic sum and Taylor's polynomial,

$$\begin{aligned}\mathcal{H}_{pq}^{(M-i)} - \mathcal{H}_{pq}^{(M)} = -(\mathcal{H}_{pq}^{(M)} - \mathcal{H}_{pq}^{(M-i)}) &= -\sum_{j=M-i}^{M-1} (\mathcal{H}_{pq}(q, p_{j+1}) - \mathcal{H}_{pq}(q, p_j)) = \\ &= -\sum_{j=M-i}^{M-1} (\mathcal{H}_{pq}(q, p_j) + \mathcal{O}(p_{j+1} - p_j) - \mathcal{H}_{pq}(q, p_j)) = \\ &= -\sum_{j=M-i}^{M-1} \mathcal{O}(h^{j+1}) = \mathcal{O}(h^{M-i+1}).\end{aligned}$$

We plug this expression into (4.3), which yields

$$\Delta_M = \sum_{i=1}^{M} (-h)^i (\mathcal{H}_{pq}^{(M-i)} - \mathcal{H}_{pq}^{(M)}) \prod_{j=1}^{i-1} \mathcal{H}_{pq}^{(M-j)} = \sum_{i=1}^{M} h^i \mathcal{O}(h^{M-i+1}) = \mathcal{O}(h^{M+1}).$$

Therefore, putting the previous equation for Δ_M into (4.2),

$$\iota_M = 1 - (-h)^{M+1} \prod_{j=0}^{M} \mathcal{H}_{pq}^{(j)} + \Delta_M = 1 + \mathcal{O}(h^{M+1}) + \mathcal{O}(h^{M+1}) = 1 + \mathcal{O}(h^{M+1}).$$

From the discussion at the beginning of this section, we finally have

$$J_{\text{new}} = \begin{pmatrix} 0 & \iota_M \\ -\iota_M & 0 \end{pmatrix} = \iota_M J = (1 + \mathcal{O}(h^{M+1}))J.$$

□

4.1.1 Optimality of the estimate

Finally, let us show that we cannot generally improve the bound from Theorem 10. Let

$$\mathcal{H} = \frac{1}{2}q^2 + qp + \frac{1}{2}p^2.$$

Then, $\mathcal{H}_q = q + p = \mathcal{H}_p$. We can show by mathematical induction that

$$\partial_p p_n = \sum_{i=0}^{n} (-h)^i$$

—for $n = 0$ we have $\partial_p p_0 = \partial_p p = 1$ and if the $\partial_p p_n$ expression holds for n, then

$$\partial_p p_{n+1} = \partial_p (p - h(q + p_n)) = 1 - h\partial_p p_n = 1 - h \sum_{i=0}^{n} (-h)^i = \sum_{i=0}^{n+1} (-h)^i.$$

Therefore (for this Hamiltonian $\mathcal{H}_{pq} = 1$),

$$\iota_M = (1+h\cdot\mathcal{H}_{pq}^{(M)})\partial_p p_M = (1+h\cdot 1) \sum_{i=0}^{M} (-h)^i = (1+h)\frac{1 - (-h)^{M+1}}{1 - (-h)} = 1-(-h)^{M+1},$$

which shows that for this particular Hamiltonian, we get a perturbation of the symplecticity

$$J_{\text{new}} = \iota_M J = (1 - (-h)^{M+1})J,$$

so the estimate $J_{\text{new}} = (1 + \mathcal{O}(h^{M+1}))J$ cannot be in general improved.

4.1.2 Comparison with a similar result

The way how iterative root finding algorithm affects the symplecticity condition is only sparsely studied in the literature. A similar result is presented in [13], where a general symplectic Runge-Kutta method combined with the FPI method is employed, with the same initial approximation as our $p_0 := p$ (in the paper, the previous step is z_0 and the fixed point iteration is $y^{[0]} := z_0$, $y^{[n+1]}$ is then always computed by FPI from $y^{[n]}$ and the resulting new step is calculated from $y^{[N]}$ for a fixed $N \in \mathbb{N}$). The paper's variables N and τ correspond to the variables M and h in this thesis and we shall keep using these.

The paper essentially proves that for $\delta := Ch$, where $C > 0$ is a constant given only by the RK method and the studied problem, the symplecticity error satisfies

$$\left\| \left(D\Phi_h^{[M]}\right)^T J \left(D\Phi_h^{[M]}\right) - J \right\|_2 = \mathcal{O}\Big((1+M)\delta^{M+2}\Big), \qquad h \to 0, \tag{4.4}$$

where the norm is the spectral matrix norm

$$\|A\|_2 := \sup\{\|Ax\| \mid \|x\| = 1\}.$$

The work [13] moreover estimates the actual constants which are hidden in the $\mathcal{O}$-notation.

This result given in [13] is more general than the result presented in the previous section in the sense that it assumes a whole class of symplectic integrators (not just the Symplectic Euler method) in a general $\mathbb{R}^{2N}$ space (not just $\mathbb{R}^2$). However, Theorem 10 is not just a direct consequence of (4.4), because the Symplectic Euler method is not a Runge-Kutta method, but rather a partitioned Runge-Kutta method, which is a different class of numerical integrators.

Moreover, (4.4) estimates the difference between the matrices J_{new} and J with respect to the matrix distance $\| \cdot - \cdot \|_2$, which does not give much information about the matrix structure of J_{new}. On the other hand, Theorem 10 shows that in the case of $\mathbb{R}^2$, the matrix J_{new} has a very similar structure as the original matrix J, as it is just J multiplied by a scalar. In the next section, we discuss the matrix structure of J_{new} for the general case $\mathbb{R}^{2N}$, $N \geq 2$.

Finally, suppose that the estimates given by (4.4) and Theorem 10 could be directly compared (e.g. the result from [13] would also hold for partitioned Runge-Kutta methods and Theorem 10 would give the same asymptotic estimate[1] for the spectral norm). Firstly, (4.4) exponentially scales with $\delta = Ch$ instead of h, which can be a worse or better estimate than Theorem 10 depending on whether $C > 1$ or $0 < C < 1$. Secondly, assume that $C = 1$, then (4.4) essentially scales as $\mathcal{O}(Mh^{M+2})$, whereas Theorem 10 scales as $\mathcal{O}(h^{M+1})$. For M large enough, $Mh > 1$ and the asymptotic behaviour of the estimate $\mathcal{O}(Mh^{M+2})$ is worse than of the estimate $\mathcal{O}(h^{M+1})$. However, for smaller M (which is relevant for practical purposes), the estimate $\mathcal{O}(h^{M+1})$ is worse, provided that the $\mathcal{O}$-hidden constants in both estimates are somewhat similar. For cases $C \neq 1$, the linear M-dependency in $\mathcal{O}(M\delta^{M+2})$ is irrelevant for this asymptotic comparison.

[1] This assumption is valid. It always holds $\| \cdot \|_2 \leq \| \cdot \|_F$, where $\| \cdot \|_F$ denotes the Frobenius norm, so if $J_{\text{new}} = \iota_M J$, $\iota_M > 0$, then $\|J_{\text{new}} - J\|_2 \leq \|J_{\text{new}} - J\|_F = |\iota_M - 1|\sqrt{2} = \mathcal{O}(h^{M+1})$, where $\sqrt{2}$ is due to $J \in \mathbb{R}^{2\times 2}$.

4.2 Symplecticity criterion and matrix structure

For a numerical flow Φ_h of a general numerical integrator, the matrix of interest $(D\Phi_h)^T J (D\Phi_h)$ can be rewritten as

$$(D\Phi_h)^T J(D\Phi_h) = \begin{pmatrix} \tilde{q}_q & \tilde{q}_p \\ \tilde{p}_q & \tilde{p}_p \end{pmatrix}^T \begin{pmatrix} O & I \\ -I & O \end{pmatrix} \begin{pmatrix} \tilde{q}_q & \tilde{q}_p \\ \tilde{p}_q & \tilde{p}_p \end{pmatrix} = \begin{pmatrix} \tilde{q}_q^T & \tilde{p}_q^T \\ \tilde{q}_p^T & \tilde{p}_p^T \end{pmatrix} \begin{pmatrix} \tilde{p}_q & \tilde{p}_p \\ -\tilde{q}_q & -\tilde{q}_p \end{pmatrix},$$

so

$$(D\Phi_h)^T J(D\Phi_h) = \begin{pmatrix} \tilde{q}_q^T \tilde{p}_q - \tilde{p}_q^T \tilde{q}_q & \tilde{q}_q^T \tilde{p}_p - \tilde{p}_q^T \tilde{q}_p \\ \tilde{q}_p^T \tilde{p}_q - \tilde{p}_p^T \tilde{q}_q & \tilde{q}_p^T \tilde{p}_p - \tilde{p}_p^T \tilde{q}_p \end{pmatrix}. \tag{4.5}$$

This matrix has a special structure: If we denote $Q := \tilde{q}_q^T \tilde{p}_p - \tilde{p}_q^T \tilde{q}_p$ and define

$$[A, B] := A^T B - B^T A,$$

then

$$(D_h\Phi)^T J(D_h\Phi) = \begin{pmatrix} [\tilde{q}_q, \tilde{p}_q] & Q \\ -Q^T & [\tilde{q}_p, \tilde{p}_p] \end{pmatrix}. \tag{4.6}$$

In the previous text, we have already indirectly exploited this structure:

1. When proving that the Symplectic Euler scheme (3.1) is symplectic, we showed $Q = I$, which meant that $-Q^T = -I$, and thus we did not have to explicitly compute the block in the second row and the first column of the resulting matrix $(D_h\Phi)^T J(D_h\Phi)$;

2. In the previous analysis of the FPI-modified Symplectic Euler scheme (4.1), we assumed the case $N = 1$, for which any $A, B \in \mathbb{R}^{N\times N}$ correspond to scalars, so $[A, B] = O = (0)$, i.e. the diagonal of $(D_h\Phi)^T J(D_h\Phi)$ was automatically zero.

However, for $N \geq 2$ and a symplectic scheme utilising fixed point iterations, the diagonal blocks are no longer necessarily zero matrices, as we shall show in the following text, again for the scheme (4.1).

From the definition of the scheme,

$$\tilde{q}_q = I + h\mathcal{H}_{qp}(q, \tilde{p}) + h\mathcal{H}_{pp}(q, \tilde{p})\tilde{p}_q. \tag{4.7}$$

Therefore, using the symmetry of the matrix $\mathcal{H}_{pp}$,

$$\begin{aligned} [\tilde{q}_q, \tilde{p}_q] &= (I + h\mathcal{H}_{qp}(q, \tilde{p}) + h\mathcal{H}_{pp}(q, \tilde{p})\tilde{p}_q)^T \tilde{p}_q - \\ &\quad - \tilde{p}_q^T (I + h\mathcal{H}_{qp}(q, \tilde{p}) + h\mathcal{H}_{pp}(q, \tilde{p})\tilde{p}_q) = \\ &= (\tilde{p}_q - \tilde{p}_q^T) + h(\mathcal{H}_{qp}(q, \tilde{p})^T \tilde{p}_q - \tilde{p}_q^T \mathcal{H}_{qp}(q, \tilde{p})). \end{aligned} \tag{4.8}$$

Both parentheses in the last equation measure the “asymmetry” of the matrices $\tilde{p}_q$ and $\mathcal{H}_{qp}(q, \tilde{p})^T \tilde{p}_q$, respectively—the corresponding difference in parentheses is zero if and only if the corresponding matrix is symmetric.

Let us analyse the simplest case possible with $M = 1$, i.e.

$$\tilde{p} = p - h\mathcal{H}_q(q, p).$$

Then, $\tilde{p}_q = -h\mathcal{H}_{qq}(q,p)$, which is a symmetric matrix, so the term $\tilde{p}_q - \tilde{p}_q^T$ in (4.8) vanishes for any smooth Hamiltonian $\mathcal{H}$. Therefore, in order to have nonzero $[\tilde{q}_q, \tilde{p}_q]$, it suffices to choose $\mathcal{H}$ so that the matrix $\mathcal{H}_{qp}(q,\tilde{p})^T\tilde{p}_q$ is not symmetric.

A simple example of such a Hamiltonian is

$$\mathcal{H}(q,p) := \frac{1}{2}\sum_{i=1}^{N} q_i^2 p_{N-i+1}.$$

(Note that contrary to the previous section, where momentum subscript p_n denoted the n-th approximation of $\tilde{p}$, here subscripts of q and p indicate the individual components of vectors $q, p \in \mathbb{R}^N$. This is an unfortunate abuse of notation, but we do not need to denote the approximation step here, as we did not need to denote the momenta components in the previous section, so the following text should not contain any ambiguous notation. Also, contrary to the physical notation of Chapter 2 using superscript q^i to denote the i-th position coordinate, we use subscript q_i here, as the superscript is already used to denote the second power.)

For this Hamiltonian, we have

$$\mathcal{H}_{qp} = \begin{pmatrix} & & & q_N \\ & & q_{N-1} & \\ & \iddots & & \\ q_1 & & & \end{pmatrix}, \qquad \mathcal{H}_{qq} = \begin{pmatrix} p_N & & & \\ & p_{N-1} & & \\ & & \ddots & \\ & & & p_1 \end{pmatrix}$$

(empty entries in matrices are zero elements). Therefore, the matrix

$$\mathcal{H}_{qp}(q,\tilde{p})^T\tilde{p}_q = -h\mathcal{H}_{qp}(q,\tilde{p})^T\mathcal{H}_{qq}(q,p) = -h\begin{pmatrix} & & & q_1p_1 \\ & & q_2p_2 & \\ & \iddots & & \\ q_Np_N & & & \end{pmatrix}$$

is in general not symmetric, and

$$\begin{aligned} [\tilde{q}_q, \tilde{p}_q] &= O + h(\mathcal{H}_{qp}(q,\tilde{p})^T\tilde{p}_q - (\mathcal{H}_{qp}(q,\tilde{p})^T\tilde{p}_q)^T) = \\ &= h^2\begin{pmatrix} & & & q_Np_N - q_1p_1 \\ & & q_{N-1}p_{N-1} - q_2p_2 & \\ & \iddots & & \\ q_1p_1 - q_Np_N & & & \end{pmatrix} \end{aligned}$$

is in general nonzero, which shows that the diagonal block in (4.6) does not have to be zero for $N \geq 2$ and symplectic schemes whose implicit equations are solved via the fixed point iteration method.

However, for the particular modification (4.1) of Symplectic Euler's method, we also have

$$\tilde{q}_p = h\mathcal{H}_{pp}(q,\tilde{p})\tilde{p}_p,$$

so

$$[\tilde{q}_p, \tilde{p}_p] = h(\mathcal{H}_{pp}(q,\tilde{p})\tilde{p}_p)^T\tilde{p}_p - \tilde{p}_p^T(h\mathcal{H}_{pp}(q,\tilde{p})\tilde{p}_p) = O$$

for arbitrary M. This shows that the scheme (4.1), employing FPI method to find $\tilde{p}$, has the last block in the matrix (4.6) equal to zero independently on the FPI approximation quality, so not all the structure of the original Symplectic Euler's method is lost.

5 Numerical experiments

In this section, we will finally employ symplectic integrators to numerically solve Hamiltonian problems.

In the first part, a simple example of a charged particle in a homogeneous electromagnetic field will be discussed. An analytical solution to that problem will also be presented, so that the numerical solution can be directly compared with the exact solution. Finally, the order of the Störmer-Verlet scheme will be empirically estimated in this first part.

Then, in the second part, we will work with a simplified model of the magnetic field in tokamak. A *tokamak* is a device for maintaining and heating of plasma (a ionised gas susceptible to electromagnetic fields) in order to study the plasma's properties or perform thermonuclear fusion. In a tokamak, the plasma can flow in a toroidal chamber, its flow being constrained by the magnetic field and accelerated by the electric field. In this second part, the magnetic field lines in a tokamak will be numerically approximated, utilising their (at first sight hidden) Hamiltonian structure [14].

We will again use more physical notation in this chapter. Namely, vector quantities will be written bold and time derivatives will be denoted by a dot over a letter.

Let us briefly present the main quantities from the theory of classical electromagnetism. For more complete discussion, see e.g. Griffiths' introductory textbook [15]. The particular Lagrangian and Hamiltonian and gauge transformations are also discussed in [9].

A mass particle can have nonzero *charge* Q, a quantity determining how the particle interacts with electric and magnetic fields. Those fields are described by vector fields $\boldsymbol{E}$ and $\boldsymbol{B}$, respectively.

Charged particles and their motions are not only affected by those fields, but they also induce them. However, we shall stay with the simplest description, where the induced fields due to our studied charged particles are negligible, so the electric and magnetic fields are given externally and they only dictate the particles' motion. This interaction for a single particle is described by the *Lorenz force*

$$\boldsymbol{F}_{\mathrm{elmg}} = Q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}),$$

where $\boldsymbol{v}$ is the particle's velocity.

The fields $\boldsymbol{E}$ and $\boldsymbol{B}$ are special in the sense that they can be written as

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}, \qquad \boldsymbol{E} = -\nabla\phi - \frac{\partial \boldsymbol{A}}{\partial t}, \tag{5.1}$$

where the scalar function ϕ is the *electric potential* and the vector function $\boldsymbol{A}$ is the *vector potential*. These potentials are not determined uniquely, as any transformation

$$\boldsymbol{A} \mapsto \boldsymbol{A} + \nabla\xi, \qquad \phi \mapsto \phi - \frac{\partial \xi}{\partial t} \tag{5.2}$$

for arbitrary smooth scalar function ξ yields the same fields in the relations (5.1). This is known as the *electromagnetic gauge transformation* (and the invariance is the *electromagnetic gauge symmetry*), which, for example, effectively lets us zero out one component of $\boldsymbol{A}$.

Using the electromagnetic potentials, the dynamics of a charged particle in an electromagnetic field can be described both in the Lagrangian and Hamiltonian formalism. The Lagrangian is given by

$$\mathcal{L}_{\text{elmg}} = \frac{1}{2}mv^2 + Q\boldsymbol{v}\cdot\boldsymbol{A} - Q\phi \tag{5.3}$$

(as we have already mentioned in Section 2.1.1) with $v := \|\boldsymbol{v}\|$. The Hamiltonian has the form

$$\mathcal{H}_{\text{elmg}} = \frac{1}{2m}\|\boldsymbol{p} - Q\boldsymbol{A}\|^2 + Q\phi. \tag{5.4}$$

5.1 Charged particle in a homogeneous electromagnetic field

The problem is to describe the motion of a particle with mass m and charge Q, which is subject to homogeneous perpendicular electric and magnetic fields. This problem is also solved and discussed in [14], Section 1.1.4, for a bit more restrictive initial condition $\boldsymbol{v}_0 = \boldsymbol{0}$.

We choose the Cartesian coordinate frame, in which the electric field is oriented parallel to the x-axis, the magnetic field is oriented parallel to the z-axis and the particle is initially located in the origin of the coordinate frame with a general initial velocity $\boldsymbol{v}_0 = (v_x, v_y, v_z)^T$.

This choice means that the fields are given by

$$\boldsymbol{E} = (E, 0, 0)^T, \qquad \boldsymbol{B} = (0, 0, B)^T,$$

where E and B are constants. If we choose $\boldsymbol{A} = (0, Bx, 0)^T$, the first equation in (5.1) is satisfied and we get the correct magnetic field. Moreover, this vector potential is time-independent, so in order to satisfy the second equation in (5.1), it suffices to choose $\phi = -Ex$.

The Hamiltonian is then given by

$$\mathcal{H} = \frac{1}{2m}\|\boldsymbol{p} - Q\boldsymbol{A}\|^2 + Q\phi = \frac{1}{2m}(p_x^2 + (p_y - QBx)^2 + p_z^2) - QEx,$$

so

$$\nabla_q\mathcal{H} = \left(-\frac{QB}{m}(p_y - QBx) - QE, 0, 0\right)^T, \quad \nabla_p\mathcal{H} = \left(\frac{p_x}{m}, \frac{1}{m}(p_y - QBx), \frac{p_z}{m}\right)^T,$$

where $q = (x, y, z)^T$ and $p = (p_x, p_y, p_z)^T$.

5.1.1 Analytical solution

The Hamilton's equations (2.3) are in this case

$$\begin{aligned}
\dot{x} &= \frac{p_x}{m}, & \dot{p}_x &= \frac{QB}{m}(p_y - QBx) + QE,\\
\dot{y} &= \frac{1}{m}(p_y - QBx), & \dot{p}_y &= 0,\\
\dot{z} &= \frac{p_z}{m}, & \dot{p}_z &= 0,
\end{aligned}$$

which means that $p_y \equiv P_y = \text{const}$ and $p_z \equiv P_z = \text{const}$, and so the equation $\dot{z} = p_z/m$ has the solution

$$z(t) = \frac{P_z}{m} t = v_z t,$$

as we chose $z(0) = 0$. The choice $v_z = P_z/m$ satisfies the condition $\dot{z}(0) = v_z$.

Now we take the equation $\dot{x} = p_x/m$, differentiate its both sides and substitute into it the equation for $\dot{p}_x$, which yields

$$\ddot{x} = \frac{\dot{p}_x}{m} = \frac{QB}{m^2}(P_y - QBx) + \frac{QE}{m},$$

so

$$\ddot{x} + \left(\frac{QB}{m}\right)^2 x = \frac{QE}{m} + \frac{QB}{m^2} P_y.$$

Denoting $\omega := QB/m$, we have

$$\ddot{x} + \omega^2 x = \frac{QE}{m} + \frac{QB}{m^2} P_y,$$

which is the equation of a harmonic oscillator with constant right hand side.

This equation has the solution

$$x(t) = \alpha \cos \omega t + \beta \sin \omega t + \frac{1}{\omega^2}\left(\frac{QE}{m} + \frac{QB}{m^2} P_y\right) = \alpha \cos \omega t + \beta \sin \omega t + \frac{Em}{QB^2} + \frac{P_y}{QB},$$

which yields the velocity

$$\dot{x}(t) = \beta\omega \cos \omega t - \alpha\omega \sin \omega t.$$

Setting $\dot{x}(0) = v_x$, we get $\beta = v_x/\omega$. Denoting

$$R := \frac{Em}{QB^2} + \frac{P_y}{QB}$$

and using the condition $x(0) = 0$, we also get the formula $\alpha = -R$, so that

$$x(t) = R(1 - \cos \omega t) + \frac{v_x}{\omega} \sin \omega t.$$

Finally, from the last unused Hamilton's equation,

$$\dot{y} = \frac{1}{m}(P_y - QBx),$$

so by integrating the right hand side with respect to time, we get

$$y(t) = \frac{P_y}{m} t - \frac{QB}{m}\left(Rt - \frac{R}{\omega} \sin \omega t - \frac{v_x}{\omega^2} \cos \omega t\right) - \frac{v_x}{\omega}$$

(the last term is an integration constant so that $y(0) = 0$). Because

$$\frac{QB}{m} R = \frac{E}{B} + \frac{P_y}{m}, \qquad \frac{QB}{m} = \omega,$$

we finally have

$$y(t) = R \sin \omega t + \frac{v_x}{\omega} \cos \omega t - \frac{v_x}{\omega} - \frac{E}{B} t.$$

Moreover, we get

$$\dot{y}(t) = R\omega \cos \omega t - v_x \sin \omega t - \frac{E}{B},$$

so

$$v_y = \dot{y}(0) = R\omega - \frac{E}{B} = \frac{P_y}{m}.$$

Altogether, the particle moves with a constant speed in the z-direction and the trajectory in x and y directions resembles a trochoid [14].

5.1.2 Numerical solution

In order to solve the problem numerically, we employ the Symplectic Euler scheme (3.1) and (nonsymplectic) Explicit Euler scheme (1.3).

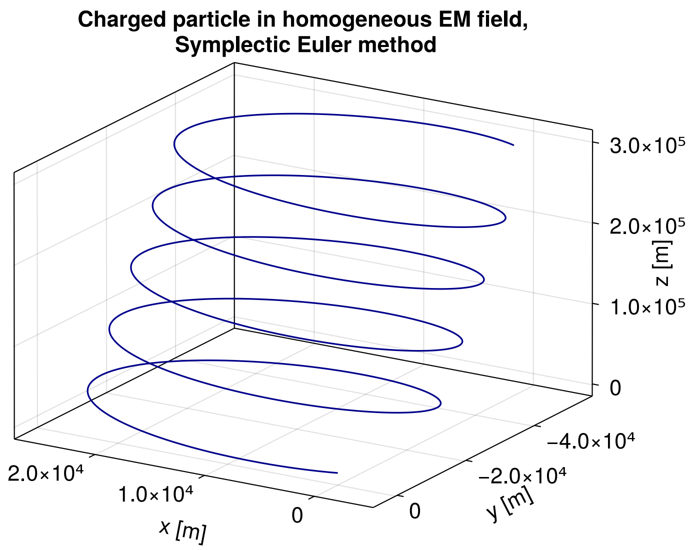

Figure 5.1 Charged particle's trajectory in a homogeneous electromagnetic field. The physical parameters are: $Q = 10^{-19}\,\mathrm{C}$, $m = 10^{-27}\,\mathrm{kg}$, $B = 0.1\,\mathrm{T}$, $E = 1\,\mathrm{N/C}$, $\boldsymbol{v}_0 = (100, 100, 100)^T\,\mathrm{m/s}$, $\boldsymbol{x}_0 = (0, 0, 0)^T\,\mathrm{m}$. The integrator is Symplectic Euler method with step size $h = 0.1\,\mathrm{s}$ and number of steps $N_{\mathrm{steps}} = 3 \cdot 10^4$.

The Symplectic Euler scheme gives these recurrences (with p being the old momentum and $\tilde{p}$ the new momentum):

$$\begin{aligned}
\tilde{p}_y &= p_y, \\
\tilde{p}_z &= p_z, \\
\tilde{p}_x &= p_x + h\left(QE + \frac{QB}{m}(\tilde{p}_y - QBx)\right), \\
\tilde{x} &= x + h\frac{\tilde{p}_x}{m}, \\
\tilde{y} &= y + \frac{h}{m}\left(\tilde{p}_y - QBx\right), \\
\tilde{z} &= z + h\frac{\tilde{p}_z}{m}.
\end{aligned}$$

Note that in this example, the recurrences are actually explicit, because the vector potential $\boldsymbol{A}$ has a simple form. Moreover, only four from the six state coordinates need to be updated (p_y and p_z are constant).

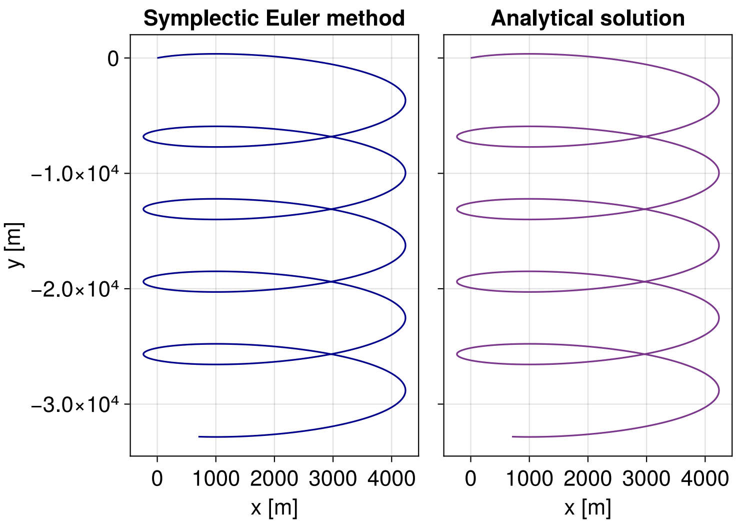

Figure 5.2 Charged particle's trajectory in a homogeneous electromagnetic field. The parameters differ from the previous Figure 5.1 only in the initial velocity, which is now $\boldsymbol{v}_0 = (10, 10, 0)^T$ m/s. The Symplectic Euler solution is visually in agreement with the analytical solution derived in the previous section.

Figure 5.1 shows a particle's trajectory in the studied physical system. The parameters are chosen (not exactly, just the order of magnitude) so that the system corresponds to a proton in a strong magnetic field. In this case, the initial velocity's z-component was nonzero, so the resulting trajectory is three-dimensional. The motion in the z-direction is, however, very simple, and if the z-component of $\boldsymbol{v}_0$ is zero, the motion is completely planar, which admits a somewhat simpler visualisation, as in Figure 5.2.

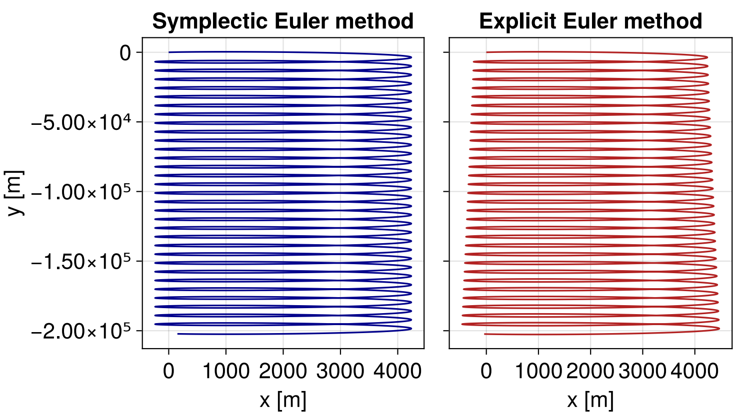

Figure 5.3 Charged particle's trajectory in a homogeneous electromagnetic field. The parameters are the same as in the previous Figure 5.2, but the number of steps is $N_{\text{steps}} = 2 \cdot 10^5$. The Explicit and Symplectic Euler methods are compared.

In Figure 5.3, trajectories found using both Explicit and Symplectic Euler methods are compared over a longer time interval. We notice that for the Explicit Euler method, the cycles are getting progressively larger, whereas for the Symplectic Euler method, their size does not visually change. This effect affects both the global errors of the methods (which cannot be in general computed,

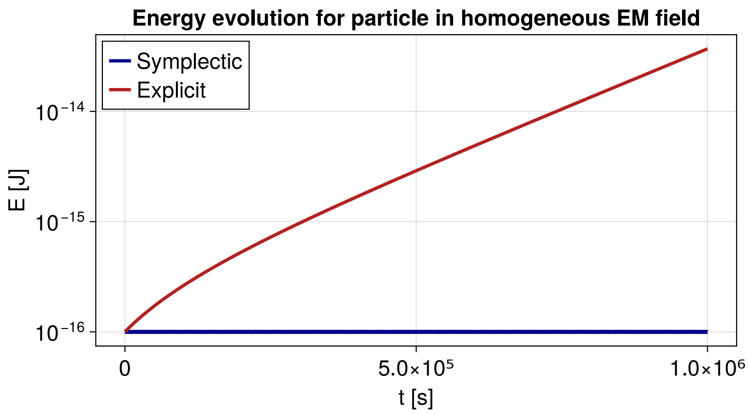

Figure 5.4 Particle's generalised energy (Hamiltonian) with respect to time. Symplectic and Explicit Euler methods are compared. The number of steps was changed to $N_{\text{steps}} = 2 \cdot 10^7$ and the step size to $h = 0.05\,\text{s}$.

but in this case, the analytical solution is a priori known) and the conservation of the Hamiltonian (which can be always computed), as the Hamiltonian is time-independent and we can use Theorem 2.

The evolution of the Hamiltonian/generalised energy with respect to time is visualised in Figure 5.4. We see that the energy blows up for the Explicit Euler method, whereas the Symplectic Euler method stays approximately on the same constant energy level. This is anticipated behaviour, as the symplectic integrators are known (both theoretically and empirically) to conserve the system's energy well.

5.1.3 Estimating the order of the Störmer-Verlet scheme

The knowledge of the analytical solution for our problem lets us compute global errors of an integrator directly, which then can be used to empirically estimate the order of the particular integrator. This approach is somewhat similar to the one employed in [3], Chapter II, Section 4, where the constant of the leading error term is estimated instead.

Let $(e_i^{[h]})_{i=0}^{n(h)}$ be the global errors for a fixed numerical scheme with step size h integrating over a fixed time interval $[0, \tau]$ (i.e. $h \cdot n(h) = \tau$). Let

$$\epsilon_h := \max_{0 \le i \le n(h)} |e_i^{[h]}|$$

be the largest error over that time interval. The idea is to assume that approximately,

$$\epsilon_h \approx C h^p, \tag{5.5}$$

where C is a constant independent on h and p is the order of the numerical method. This is motivated by the fact that under mild assumptions, a scheme of order p at least satisfies $\epsilon_h = \mathcal{O}(h^p)$, $h \to 0$. If we then use two different step sizes h_1, h_2, we have

$$\epsilon_{h_1} \approx C h_1^p, \qquad \epsilon_{h_2} \approx C h_2^p,$$

so

$$\frac{\epsilon_{h_1}}{\epsilon_{h_2}} \approx \left(\frac{h_1}{h_2}\right)^p,$$
$$\log\left(\frac{\epsilon_{h_1}}{\epsilon_{h_2}}\right) \approx p \log\left(\frac{h_1}{h_2}\right),$$

and therefore

$$p \approx \frac{\log \epsilon_{h_1} - \log \epsilon_{h_2}}{\log h_1 - \log h_2} =: \frac{\Delta \log \epsilon_{h_i}}{\Delta \log h_i}. \tag{5.6}$$

The integrator's order can be thus estimated by measuring how the change of step size h affects the max-error ϵ_h.

We shall try this for the Störmer-Verlet scheme (3.9), which is known to have order 2 (but we have not proven it in this work). The Störmer-Verlet scheme gives the following recurrences for this problem (again, p is the old momentum, $\tilde{p}$ is the new momentum, but this time there is also the intermediate momentum $\overline{p}$, corresponding to $p_{n+1/2}$ in (3.9))

$$\begin{aligned}
\tilde{p}_y &= \overline{p}_y = p_y, \\
\tilde{p}_z &= \overline{p}_z = p_z, \\
\overline{p}_x &= p_x + \frac{h}{2}QE + \frac{h}{2}\frac{QB}{m}(\overline{p}_y - QBx), \\
\tilde{x} &= x + h\frac{\overline{p}_x}{m}, \\
\tilde{y} &= y + \frac{h}{2m}(\overline{p}_y - QBx + \overline{p}_y - QB\tilde{x}), \\
\tilde{z} &= z + h\frac{\overline{p}_z}{m}, \\
\tilde{p}_x &= \overline{p}_x + \frac{h}{2}QE + \frac{h}{2}\frac{QB}{m}(\overline{p}_y - QB\tilde{x}).
\end{aligned}$$

Just as in the implementation of the Symplectic Euler method for this problem, the values p_y and p_z are constant along the computation and each step can be computed explicitly.

h [s]	ϵ_h [m]	p_{estim}
1	6.3234	—
0.1	$6.3234 \cdot 10^{-2}$	2
0.05	$1.5808 \cdot 10^{-2}$	2.0000456
0.01	$6.3234 \cdot 10^{-4}$	1.9999803
0.005	$1.5808 \cdot 10^{-4}$	2.0000456

Table 5.1 Order estimation for the Störmer-Verlet scheme. The right column is always computed from the actual and previous row using the formula (5.6). The physical parameters are the same as in Fig. 5.1, the total integration time is held fixed at 10000 s.

The Table 5.1 shows that empirically, the order of the Störmer-Verlet scheme indeed appears to be 2. For smaller h values, however, the error ϵ_h stops decreasing according to the assumption (5.5). This could be due to the rounding errors in the finite precision arithmetic and it is a limitation of this order estimation approach.

5.2 Tokamak Field Lines

In this second example, we shall numerically calculate *magnetic field lines* of a known magnetic field. The magnetic field lines are just integral curves for the given field, i.e. parametric curves $\boldsymbol{c}$ for which at every parameter value λ, the tangent vector of $\boldsymbol{c}$ is determined by the (time-independent) magnetic field $\boldsymbol{B}$ as

$$\boldsymbol{c}'(\lambda) = \boldsymbol{B}(\boldsymbol{c}(\lambda)). \tag{5.7}$$

The field lines are useful in plasma physics, because although their description is very simple, they carry relevant information. A charged particle in a magnetic field tends to circulate around a field line, so if the radius of circulation[1] is small enough and the magnetic field does not change too much in space, the field lines give a good idea about the dynamics of charged particles in that magnetic field. Moreover, the field lines describe manifolds of approximately constant pressure and which constrain the equilibrium currents, which is useful on its own [14, 16].

From a numerical point of view, field lines:

1. Are easier to integrate. We shall see in the following text that the problem can be reformulated as a mechanical problem with one degree of freedom, opposed to the general three degrees of freedom for a charged particle.
2. In general admit a larger integration step size—because charged particles tend to circulate the field lines, the step size needs to be small enough so that these fast circulations are actually represented in the discretisation, even though these individual circulations are not that interesting on their own.

Finally, the problem is relevant for this thesis because it admits a Hamiltonian formulation.

5.2.1 Hamiltonian formalism of magnetic field lines

We follow here the derivation of Hamilton's equations for magnetic field lines as it is described in [16]. The magnetic field lines formalism builds upon this single postulate:

Let $\boldsymbol{A}$ be a vector potential of the given magnetic field $\boldsymbol{B}$. A curve $\boldsymbol{c}$ is a magnetic field line of the field $\boldsymbol{B}$ connecting points $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ if and only if $\boldsymbol{c}$ is a stationary point of the functional

$$\mathcal{F} : \boldsymbol{g} \mapsto \int_{\boldsymbol{g}} \boldsymbol{A}(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{x},$$

where $\boldsymbol{g}$ is a general smooth curve with endpoints $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$.

This is analogous to *Hamilton's principle* (or the *principle of stationary action*), which states that the physical trajectory between points $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ and times t_1 and t_2 of a system with Lagrangian $\mathcal{L}$ is the trajectory $\boldsymbol{c}$ such that $\boldsymbol{c}(t_i) = \boldsymbol{x}_i$, $i \in \{1, 2\}$, and $\boldsymbol{c}$ is a stationary point of the *action functional*

$$\mathcal{S} : \boldsymbol{g} \mapsto \int_{t_1}^{t_2} \mathcal{L}(t, \boldsymbol{g}(t), \dot{\boldsymbol{g}}(t))\mathrm{d}t,$$

[1] More precisely, the momentary *Larmor radius* $\frac{mv_\perp}{QB}$, where $v_\perp$ is the velocity component perpendicular to the magnetic field at the particular particle's position.

where $\boldsymbol{g}$ is a general smooth curve satisfying $\boldsymbol{g}(t_i) = \boldsymbol{x}_i$ (see also [8], Part II, Chapter 3).

This is usually formulated in terms of *calculus of variations* that the variation of this functional in $\boldsymbol{c}$ is zero, i.e.

$$\delta\mathcal{S}[\boldsymbol{c}] = 0.$$

Several remarks should now be noted:

1. We do not aim to describe the variational principles of physics in great detail (even though there are interesting consequences even for the topic of this thesis—some symplectic integrators can be obtained by discretising the action functional, see [4], Chapter VI, Section 6). We only need to algebraically manipulate the evaluated functional $\mathcal{F}[\boldsymbol{c}]$ into a form resembling the evaluation $\mathcal{S}[\boldsymbol{c}]$, rewritten in equivalent way in coordinates and momenta as
$$\int_{t_1}^{t_2} \left(\sum_{i=1}^{N} p_i(t)\dot{q}^i(t) - \mathcal{H}(t, q^j(t), p_j(t))\right)\mathrm{d}t. \tag{5.8}$$
That is the whole idea—manipulate the integral $\int_{\boldsymbol{g}} \boldsymbol{A}(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{x}$ and identify which variables have the role of t, $\mathcal{H}$, q and p so that the variation of the original integral is equivalent to the variation of an integral of the form (5.8). Then, Hamilton's equations will follow as a consequence of Hamilton's principle.

2. The whole variational approach might seem disappointing and unmotivated. This is a matter of perspective, as the mathematical description of the physical world needs to start somewhere, be it forces, Hamiltonian formalism or the stationary action. From Hamilton's principle, all of the mechanical formalisms can be derived—the Euler-Lagrange equations (2.2) actually mathematically specify when the action functional $\mathcal{S}$ is stationary, so in this sense they stem from the more general calculus of variations (and they describe physics mainly thanks to the particular choice of the functional $\mathcal{S}$ subject to variations). The Euler-Lagrange equations can then, for particular Lagrangians, yield the classical Newton's description of mechanics via forces.

3. The variational formulation for field lines can be understood as Hamilton's principle applied to the standard electromagnetic Lagrangian (5.3) in an idealised scenario of a massless charged particle in a pure magnetic field. In that case, $m = 0 = \phi$, so $\mathcal{L} = Q\boldsymbol{v} \cdot \boldsymbol{A}$. The charge $Q > 0$ is irrelevant here, so the stationary action principle can be written as
$$0 = \delta \int_{t_1}^{t_2} \mathcal{L}(t, \boldsymbol{c}(t), \dot{\boldsymbol{c}}(t))\mathrm{d}t = \delta \int_{t_1}^{t_2} \boldsymbol{A}(\boldsymbol{c}(t)) \cdot \dot{\boldsymbol{c}}(t)\mathrm{d}t = \delta \int_{\boldsymbol{c}} \boldsymbol{A}(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{x}$$
from the definition of a line integral. This is also motivated by the idea that for $m \searrow 0$, the Larmov radius should also approach zero pointwise, so the trajectories of successively lighter charged particles should approach the magnetic field line. Even without these motivations, the previous point still holds—the functional $\mathcal{F}$ is a description which yields the correct physical results, so it is useful.

We will now derive the Hamiltonian formulation of the field lines problem from the variational principles. Assume first that we use the standard Cartesian coordinates. Then, for a curve $\boldsymbol{g}$ parametrised by a parameter λ,

$$\int_{\boldsymbol{g}} \boldsymbol{A}(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{x} = \int_{\lambda_1}^{\lambda_2} \boldsymbol{A}(\boldsymbol{g}(\lambda)) \cdot \frac{\mathrm{d}\boldsymbol{g}}{\mathrm{d}\lambda}(\lambda)\mathrm{d}\lambda = \\ = \int_{\lambda_1}^{\lambda_2} \Big(A_x(\boldsymbol{g}(\lambda)) \frac{\mathrm{d}g_x}{\mathrm{d}\lambda}(\lambda) + A_y(\boldsymbol{g}(\lambda)) \frac{\mathrm{d}g_y}{\mathrm{d}\lambda}(\lambda) \Big) \mathrm{d}\lambda,$$

where we exploited the gauge freedom from gauge transformation (5.2) to zero out the vector potential's z-component A_z. Moreover, assume that it is possible to parametrise the curve using g_x or g_y (analogously to viewing some curves as functions, for example the curve $t \mapsto (t^2, t)$ can be naturally understood as $x = y^2$, which is "a parametrisation in y"). For example, assume that we can parametrise the curve as $g_x = \lambda$ and $g_y = G(\lambda)$.

Then, we have

$$\int_{\boldsymbol{g}} \boldsymbol{A}(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{x} = \int_{\lambda_1}^{\lambda_2} \Big(A_x(\boldsymbol{g}(\lambda)) + A_y(\boldsymbol{g}(\lambda)) \frac{\mathrm{d}G}{\mathrm{d}\lambda}(\lambda) \Big) \mathrm{d}\lambda.$$

If we now denote

$$t \sim \lambda, \qquad \mathcal{H} \sim -A_x, \\ p \sim A_y \circ \boldsymbol{g}, \qquad q \sim G,$$

and express $\boldsymbol{g}$ in terms of q, p and t, we finally get

$$\delta \int_{\boldsymbol{g}} \boldsymbol{A}(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{x} = \delta \int_{t_1}^{t_2} \Big(-\mathcal{H}(t, q(t), p(t)) + p(t)\dot{q}(t) \Big) \mathrm{d}t,$$

so the variational principle for field lines can be reformulated in terms of Hamilton's principle for variables corresponding to the Hamiltonian, time, momentum and position. The problem can then be restated in terms of Hamilton's equations (2.3) and solved using symplectic integrators.

We will show an application of this procedure in the next section, as a particular example is more illuminating than the general description. We just finish this general discussion by three notes: Firstly, this approach can be used with other generalised coordinates than the most simple Cartesian coordinates. We will demonstrate this with cylindrical coordinates in the next example. Secondly, solving the Hamilton's equations gives the relation $G(\lambda) \sim q(t)$, i.e. how one coordinate of the fieldline depends on its other coordinate. There is, however, one missing coordinate dependency, which often needs to be expressed from the dependency $A_y(\boldsymbol{g}(\lambda)) \sim p(t)$. Lastly, we mentioned earlier that the field line approach corresponds to a system with one degree of freedom (instead of possibly anticipated three degrees of freedom). We see that this is really the case, because the final Hamilton's equations use only one coordinate $q \sim G$ (together with the conjugate momentum $p \sim A_y \circ \boldsymbol{g}$).

5.2.2 Magnetic field lines of a simplified tokamak model

Coordinate description

In order to describe the geometry of the tokamak's torus, it is useful to use different coordinates than the plain Cartesian coordinates. The magnetic field that is employed in this section is described in [17], where a combination of cylindrical (ρ, φ, z) and toroidal (φ, θ, r) coordinates are used. For the numerical integration, we will use the cylindrical coordinates. These are well described in [15], Chapter 1 (mainly Section 4) and in Appendix A.

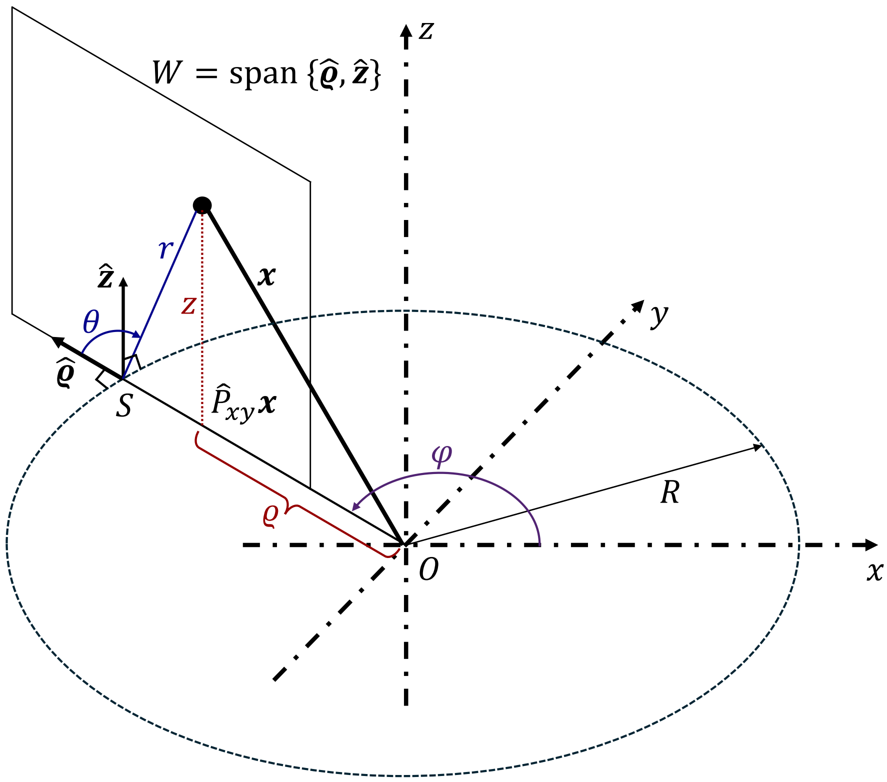

Figure 5.5 Cartesian (x, y, z) coordinates, cylindrical (ρ, φ, z) coordinates (red colour) and toroidal (φ, θ, r) coordinates (blue colour) of a point $\boldsymbol{x} \in \mathbb{R}^3$. The angle φ is coloured purple as it is used in both cylindrical and toroidal description.

The different coordinate systems are visualised in Fig. 5.5 for a general point $\boldsymbol{x} \in \mathbb{R}^3$. This point has Cartesian coordinates (x, y, z). Let $\hat{P}_{xy}\boldsymbol{x}$ denote the orthogonal projection of $\boldsymbol{x}$ onto the plane span$\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}\}$ (where $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$ is the canonical basis of $\mathbb{R}^3$). Then, define

$$\rho := \|\hat{P}_{xy}\boldsymbol{x}\| = \sqrt{x^2 + y^2}$$

and $\varphi \in [0, 2\pi)$ the angle between $\hat{\boldsymbol{x}}$ and $\hat{P}_{xy}\boldsymbol{x}$, measured counterclockwise. Then, (ρ, φ, z) are the cylindrical coordinates of the point $\boldsymbol{x}$. The angle φ is not well defined for $\rho = 0$, but in that case, the value z gives a complete information about the point's position, so we can define e.g. $\varphi := 0$ in that case. The transformation $(\rho, \varphi, z) \mapsto (x, y, z)$ is defined by

$$x = \rho \cos \varphi, \qquad y = \rho \sin \varphi, \qquad z = z. \tag{5.9}$$

At each point $\boldsymbol{x} \in \mathbb{R}^3$ for which $\rho \neq 0$, a *local cylindrical coordinate basis*

$(\hat{\boldsymbol{\rho}}(\boldsymbol{x}), \hat{\boldsymbol{\varphi}}(\boldsymbol{x}), \hat{\boldsymbol{z}}(\boldsymbol{x}))$ can be defined as

$$\begin{aligned}\hat{\boldsymbol{\rho}}(\boldsymbol{x}) &:= \frac{1}{\sqrt{x^2+y^2}}(x, y, 0)^T = \frac{1}{\rho}(\rho\cos\varphi, \rho\sin\varphi, 0)^T = (\cos\varphi, \sin\varphi, 0)^T,\\ \hat{\boldsymbol{\varphi}}(\boldsymbol{x}) &:= \frac{1}{\sqrt{x^2+y^2}}(-y, x, 0)^T = \frac{1}{\rho}(-\rho\sin\varphi, \rho\cos\varphi, 0)^T = (-\sin\varphi, \cos\varphi, 0)^T\end{aligned} \tag{5.10}$$

(and again, $\hat{\boldsymbol{z}}(\boldsymbol{x}) := (0, 0, 1)^T$, or denoted just simply $\hat{\boldsymbol{z}}$). These unit vectors define the direction in which the corresponding cylindrical coordinate changes the most, so for example, $\hat{\boldsymbol{\rho}}(\boldsymbol{x}) \propto \nabla_{(x,y,z)}\rho(\boldsymbol{x})$.

In physical systems, it is often convenient to express quantities in these local bases—it is for example often simple to express a force in the given position by its components in $\hat{\boldsymbol{\rho}}$, $\hat{\boldsymbol{\varphi}}$, $\hat{\boldsymbol{z}}$ directions. Moreover, in the case of cylindrical coordinates, these basis vectors form an orthonormal basis in each point where they are defined, which further simplifies the calculations.

The point $\boldsymbol{x} = (x, y, z)^T$ can be expressed this way as

$$\boldsymbol{x} = \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}(x, y, 0)^T + (0, 0, z)^T = \rho(\boldsymbol{x})\hat{\boldsymbol{\rho}}(\boldsymbol{x}) + z\hat{\boldsymbol{z}}. \tag{5.11}$$

If we have a curve $\boldsymbol{x} : (a, b) \to \mathbb{R}^3$ and corresponding functions $\rho : (a, b) \to (0, \infty)$ (we do not assume $\rho = 0$), $\varphi : (a, b) \to \mathbb{R}$ defining the cylindrical coordinates of the curve, then the curve's velocity at $t \in (a, b)$ is

$$\begin{aligned}\dot{\boldsymbol{x}}(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x(t)\\ y(t)\\ z(t)\end{pmatrix} = \frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\rho(t)\cos\varphi(t)\\ \rho(t)\sin\varphi(t)\\ z(t)\end{pmatrix} = \begin{pmatrix}\dot{\rho}(t)\cos\varphi(t) - \rho(t)\dot{\varphi}(t)\sin\varphi(t)\\ \dot{\rho}(t)\sin\varphi(t) + \rho(t)\dot{\varphi}(t)\cos\varphi(t)\\ \dot{z}(t)\end{pmatrix} =\\ &= \dot{\rho}(t)\begin{pmatrix}\cos\varphi(t)\\ \sin\varphi(t)\\ 0\end{pmatrix} + \rho(t)\dot{\varphi}(t)\begin{pmatrix}-\sin\varphi(t)\\ \cos\varphi(t)\\ 0\end{pmatrix} + \dot{z}(t)\begin{pmatrix}0\\ 0\\ 1\end{pmatrix},\end{aligned}$$

which yields

$$\dot{\boldsymbol{x}}(t) = \dot{\rho}(t)\hat{\boldsymbol{\rho}}(\boldsymbol{x}(t)) + \rho(t)\dot{\varphi}(t)\hat{\boldsymbol{\varphi}}(\boldsymbol{x}(t)) + \dot{z}(t)\hat{\boldsymbol{z}} \tag{5.12}$$

due to (5.10). We will use this expression while deriving Hamilton's equations for magnetic field lines in cylindrical coordinates.

Lastly, we mentioned that [17] also uses toroidal coordinates (φ, θ, r). For a point $\boldsymbol{x} \in \mathbb{R}^3$, we already know what are the vectors $\hat{\boldsymbol{\rho}}(\boldsymbol{x})$ and $\hat{\boldsymbol{z}}$ (the vector $\hat{\boldsymbol{\rho}}(\boldsymbol{x})$ is actually defined by the angle φ only, as (5.10) shows). These vectors are linearly independent, so they define a linear subspace $W := \operatorname{span}\{\hat{\boldsymbol{\rho}}(\boldsymbol{x}), \hat{\boldsymbol{z}}\}$, which is also visualised in Fig. 5.5. Because of the expression (5.11), it holds that $\boldsymbol{x} \in W$. Therefore, the angle φ determines the subspace W of dimension 2, in which $\boldsymbol{x}$ lies, and its position in this subspace can be then determined by polar coordinates (r, θ) where r is the position from a chosen point (in this setting, the point $S := R\hat{\boldsymbol{\rho}}$, where $R > 0$ is a constant, the torus' *major radius*) and θ is an angle between $\hat{\boldsymbol{\rho}}$ and $\boldsymbol{x} - S$ measured so that $\theta = \pi/2$ if $\boldsymbol{x} = S + \hat{\boldsymbol{z}}$. The angles φ and θ are called *toroidal* and *poloidal* angles, respectively.

The conversion between cylindrical and toroidal coordinates is given by the relations

$$\rho = R + r\cos\theta, \qquad r^2 = z^2 + (\rho - R)^2, \qquad z = r\sin\theta. \tag{5.13}$$

It will be useful to know that

$$2r\frac{\partial r}{\partial \rho} = 2(\rho - R) \implies \frac{\partial r}{\partial \rho} = \frac{\rho - R}{r}, \qquad 2r\frac{\partial r}{\partial z} = 2z \implies \frac{\partial r}{\partial z} = \frac{z}{r} \tag{5.14}$$

by differentiating the second expression in (5.13).

Magnetic field model in a tokamak

The magnetic field in a tokamak is often very complicated and is given not only by external coils, but also by the currents flowing inside. A somewhat simplified tokamak's magnetic field model is described in [17] and given as

$$\boldsymbol{B} = \frac{B_0}{R + r\cos\theta}(R\hat{\boldsymbol{\varphi}} + f(r)\hat{\boldsymbol{\theta}}) \tag{5.15}$$

with a corresponding vector potential

$$\boldsymbol{A} = B_0\frac{F(r)}{R + r\cos\theta}\hat{\boldsymbol{\varphi}} - B_0 R\log\left(\frac{R + r\cos\theta}{R}\right)\hat{\boldsymbol{z}} \tag{5.16}$$

(see Eq. 1 and Eq. 3 in [17]), where F is defined as $F(r) := \int_0^r f(s)\mathrm{d}s$, f is given by $f(r) = r/(qR)$ and q is chosen from Eq. 14 as

$$q(r) = \frac{1 + ar}{1 + r^2},$$

where $a > 0$ is a constant. Note that in [17], there is an extra factor R in the second term of (5.15), but this is the magnetic field consistent with the vector potential (5.16), as we now quickly check:

The curl in cylindrical coordinates is (see Vector Derivatives at the very end of [15], page 601)

$$\nabla \times \boldsymbol{A} = \left(\frac{1}{\rho}\frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z}\right)\hat{\boldsymbol{\rho}} + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right)\hat{\boldsymbol{\varphi}} + \frac{1}{\rho}\left(\frac{\partial(\rho A_\varphi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \varphi}\right)\hat{\boldsymbol{z}}.$$

In the case of (5.16), $A_\rho = 0$, $\partial_\varphi A_z = 0$ and

$$\begin{aligned}
\frac{\partial A_\varphi}{\partial z} &= \frac{B_0}{\rho}\frac{\partial}{\partial z}F(r) = \frac{B_0}{\rho}f(r)\frac{\partial r}{\partial z} = \frac{B_0}{\rho}f(r)\frac{z}{r}, \\
\frac{\partial A_z}{\partial \rho} &= -B_0 R\frac{\partial}{\partial \rho}\log\left(\frac{\rho}{R}\right) = -\frac{B_0 R}{\rho}, \\
\frac{\partial(\rho A_\varphi)}{\partial \rho} &= B_0\frac{\partial}{\partial \rho}F(r) = B_0 f(r)\frac{\partial r}{\partial \rho} = B_0 f(r)\frac{\rho - R}{r}
\end{aligned}$$

using (5.14).

Therefore,

$$\begin{aligned}
\nabla \times \boldsymbol{A} &= \left(0 - \frac{B_0}{\rho}f(r)\frac{z}{r}\right)\hat{\boldsymbol{\rho}} + \left(0 + \frac{B_0 R}{\rho}\right)\hat{\boldsymbol{\varphi}} + \frac{1}{\rho}\left(B_0 f(r)\frac{\rho - R}{r} - 0\right)\hat{\boldsymbol{z}} = \\
&= \frac{B_0}{\rho}\left(R\hat{\boldsymbol{\varphi}} + f(r)\cdot\left(\frac{\rho - R}{r}\hat{\boldsymbol{z}} - \frac{z}{r}\hat{\boldsymbol{\rho}}\right)\right) = \\
&= \frac{B_0}{\rho}\big(R\hat{\boldsymbol{\varphi}} + f(r)\cdot(\cos(\theta)\hat{\boldsymbol{z}} - \sin(\theta)\hat{\boldsymbol{\rho}})\big),
\end{aligned}$$

where we used (5.13), and which yields the desired magnetic field $\boldsymbol{B}$, because

$$\cos(\theta)\hat{\boldsymbol{z}} - \sin(\theta)\hat{\boldsymbol{\rho}} = \hat{\boldsymbol{\theta}},$$

as $\hat{\boldsymbol{\theta}}$ is given by the polar coordinates in W (in xy, it would have coordinates $(-\sin\theta, \cos\theta) = \cos(\theta)\hat{\boldsymbol{y}} - \sin(\theta)\hat{\boldsymbol{x}}$ just as $\hat{\boldsymbol{\varphi}}$ in (5.10)).

Hamilton's equations of field lines

We will again derive the Hamiltonian formulation of the field lines, this time in cylindrical coordinates. For that, note that the vector potential (5.16) is already gauged so that it has one component zero, namely the ρ-component A_ρ. Assume that a curve $\boldsymbol{c}$ can be parametrised by the angle φ and written in local cylindrical coordinates as

$$\boldsymbol{c}(\varphi) = \rho(\varphi)\hat{\boldsymbol{\rho}}(\varphi) + z(\varphi)\hat{\boldsymbol{z}}.$$

We have

$$\begin{aligned}\int_{\boldsymbol{c}} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} &= \int_{\varphi_1}^{\varphi_2} (A_\varphi(\varphi)\hat{\boldsymbol{\varphi}}(\varphi) + A_z(\varphi)\hat{\boldsymbol{z}}) \cdot \frac{\mathrm{d}\boldsymbol{c}}{\mathrm{d}\varphi}(\varphi)\mathrm{d}\varphi = \\ &= \int_{\varphi_1}^{\varphi_2} (A_\varphi(\varphi)\hat{\boldsymbol{\varphi}}(\varphi) + A_z(\varphi)\hat{\boldsymbol{z}}) \cdot (\rho'(\varphi)\hat{\boldsymbol{\rho}}(\varphi) + \rho(\varphi) \cdot 1\hat{\boldsymbol{\varphi}}(\varphi) + z'(\varphi)\hat{\boldsymbol{z}})\mathrm{d}\varphi \\ &= \int_{\varphi_1}^{\varphi_2} (A_\varphi(\varphi)\rho(\varphi) + A_z(\varphi)z'(\varphi))\mathrm{d}\varphi \sim \int_{t_1}^{t_2} (p(t)\dot{q}(t) - \mathcal{H}(t, q(t), p(t)))\mathrm{d}t,\end{aligned}$$

where we used (5.12) with the angular component satisfying $(\varphi(\varphi))' = 1$ and also the orthonormality of the local cylindrical basis.

Therefore, we get the correspondences

$$\begin{aligned} t &\sim \varphi, & \mathcal{H} &\sim -\rho A_\varphi, \\ p &\sim A_z, & q &\sim z, \end{aligned}$$

and thus the resulting Hamilton's equations

$$\frac{\mathrm{d}z}{\mathrm{d}\varphi} = -\frac{\partial(\rho A_\varphi)}{\partial A_z}, \qquad \frac{\mathrm{d}A_z}{\mathrm{d}\varphi} = \frac{\partial(\rho A_\varphi)}{\partial z}. \tag{5.17}$$

Applying this to the vector potential in (5.16), we have

$$A_z = -B_0 R \log\left(\frac{R + r\cos\theta}{R}\right) = -B_0 R \log\left(\frac{\rho}{R}\right),$$

so

$$\rho = R \exp\left(-\frac{A_z}{B_0 R}\right), \tag{5.18}$$

which defines how the missing coordinate ρ will be calculated after the system (5.17) is solved, and it also shows how does ρ depend on the "position" z, "momentum" A_z and "time" φ, which represent the actual independent variables in this formalism.

The Hamiltonian is

$$\mathcal{H} \sim -\rho A_\varphi = -\rho \cdot B_0 \frac{F(r)}{\rho} = -B_0 F(r).$$

Then,

$$\frac{\partial(-\rho A_\varphi)}{\partial A_z} = -B_0 \frac{\partial}{\partial A_z} F(r) = -B_0 f(r) \frac{\partial r}{\partial A_z} = -B_0 f(r) \frac{\partial r}{\partial \rho} \frac{\mathrm{d}\rho}{\mathrm{d}A_z},$$

because r can be expressed from (5.13) as a function of z and ρ, and z is independent on A_z. Therefore, using (5.14) and (5.18),

$$\frac{\partial(-\rho A_\varphi)}{\partial A_z} = -B_0 f(r) \frac{\rho - R}{r} R \left(-\frac{1}{B_0 R}\right) \exp\left(-\frac{A_z}{B_0 R}\right) = \frac{\rho - R}{R} \frac{\rho}{r} f(r), \qquad (5.19)$$

having the role of $\nabla_p \mathcal{H}$.

Analogously, the role of $\nabla_q \mathcal{H}$ is taken by

$$\frac{\partial(-\rho A_\varphi)}{\partial z} = -B_0 \frac{\partial}{\partial z} F(r) = -B_0 f(r) \frac{\partial r}{\partial z} = -B_0 \frac{z}{r} f(r), \qquad (5.20)$$

because r depends on ρ and z only, and ρ does not depend on z at all in (5.18).

The system (5.17) is thus in this case

$$\begin{aligned} \frac{\mathrm{d}z}{\mathrm{d}\varphi} &= \frac{\rho - R}{R} \frac{\rho}{r} f(r), \\ \frac{\mathrm{d}A_z}{\mathrm{d}\varphi} &= B_0 \frac{z}{r} f(r) \end{aligned} \qquad (5.21)$$

(where the right hand sides need to be expressed in the variables z, A_z and φ only).

An important property is that the Hamiltonian $-\rho A_\varphi = -B_0 F(r)$ is independent of φ (as r depends only on z and ρ, and ρ depends on A_z only), which implies that this Hamiltonian is an integral of motion, because φ has the role of time in this system. The quantity $-\rho A_\varphi$ is therefore conserved, which also means that F is conserved, as $-B_0$ is constant. Moreover, since f is for $R > 0$, $a > 0$ positive on $(0, \infty)$, and $F(r) = \int_0^r f(s)\mathrm{d}s$, the function F must be injective on $(0, \infty)$, and therefore the conservation of F actually implies that the field line has a constant r coordinate.

Numerical simulation

Because the field lines can be described using Hamilton's system, we can employ the Symplectic Euler method (3.1) to integrate this system. This gives the discretisation

$$\begin{aligned} \tilde{A}_z &= A_z + h B_0 \frac{z}{\tilde{r}} f(\tilde{r}), & \tilde{r} &= \sqrt{(R - \tilde{\rho})^2 + z^2}, \\ \tilde{z} &= z - h \frac{\tilde{\rho} - R}{R} \frac{\tilde{\rho}}{R} f(\tilde{r}), & \tilde{\rho} &= R \exp\left(-\frac{\tilde{A}_z}{B_0 R}\right). \end{aligned} \qquad (5.22)$$

We see that the $\tilde{A}_z$-equation in (5.22), supplemented by the equations in the second column, is strongly nonlinear for the unknown value $\tilde{A}_z$, so it needs to be solved numerically. For that, we use the fixed point iteration method, as we somewhat understand this method thanks to the Section 4 (the studied

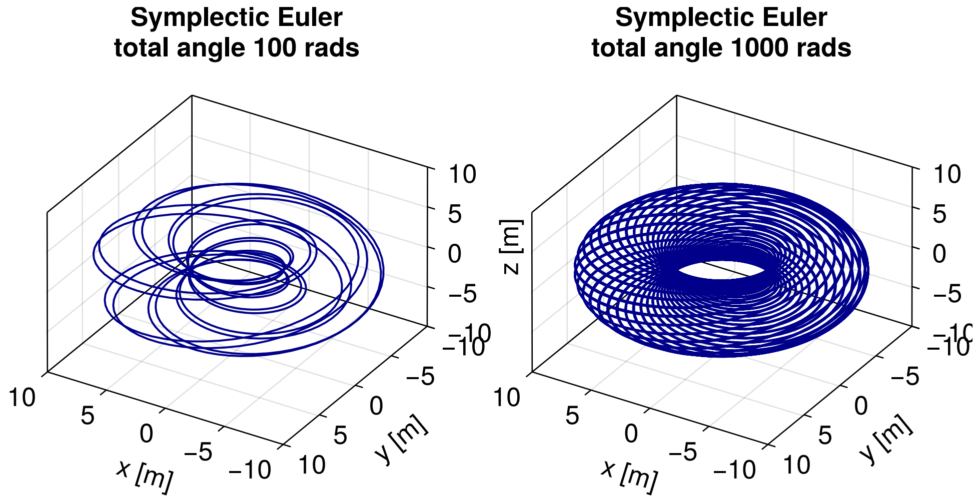

Figure 5.6 Magnetic field lines for the chosen vector potential $\boldsymbol{A}$. The parameters are $R = 7\,\mathrm{m}$, $B_0 = 5\,\mathrm{T}$, $a = 1$, $\rho_0 = 4\,\mathrm{m}$, $z_0 = -2\,\mathrm{m}$, $h = 0.01$ and 3 steps of FPI in the momentum step. The left figure uses $N_{\mathrm{steps}} = 10^4$, whereas the right figure uses $N_{\mathrm{steps}} = 10^5$.

Hamiltonian system has only one degree of freedom, so even the analysis of Section 4.1 is relevant here).

The resulting magnetic field lines are visualised in Fig. 5.6. Because the role of time is played by the toroidal angle φ, it means that if the number of steps is N_{steps} and the step size is h, then a total angle N_{steps}/h is integrated, which corresponds to $N_{steps}/(2\pi h)$ revolutions of the field line.

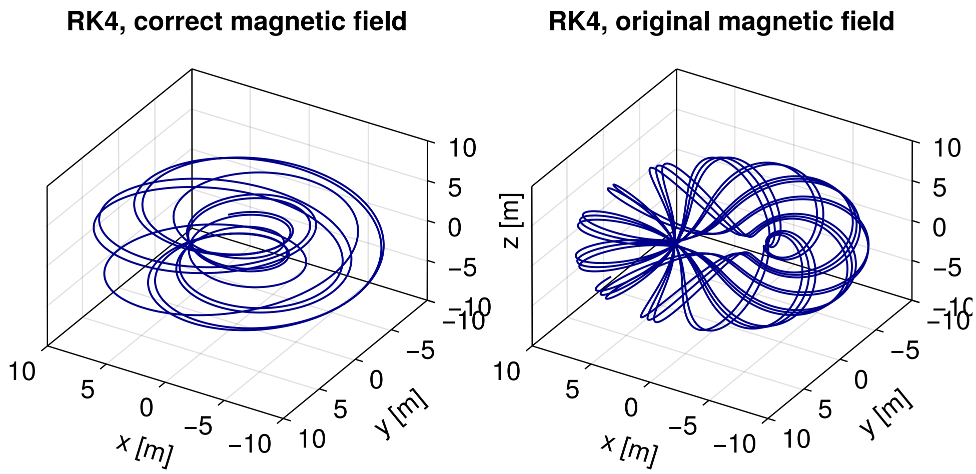

Figure 5.7 Magnetic field lines integrated by RK4 for the original and corrected magnetic fields $\boldsymbol{B}$. The parameters are $R = 7\,\mathrm{m}$, $B_0 = 5\,\mathrm{T}$, $a = 1$, $x_0 = 4\,\mathrm{m}$, $y_0 = 0\,\mathrm{m}$, $z_0 = -2\,\mathrm{m}$, $h = 0.01$ and $N_{\mathrm{steps}} = 10^4$.

We also integrate the original problem (5.7) using RK4 with the Butcher table 1.2, both for the original magnetic field from [17] and for the corrected magnetic field (5.15). This is shown in Fig. 5.7. We do this in order to check that the Hamiltonian formulation of field lines is in an agreement with a plain solution of the original problem, which indeed seems to be, contrary to the case where the original magnetic field is used (in which the field line circulates more quickly in the poloidal angle). The integration of (5.7) is performed in the Cartesian coordinates, all three of them need to be updated in each time step. The scheme is, however, explicit. The parameters N_{steps} and h are the same for Figs. 5.6 and 5.7, but the meaning of h is slightly different for these two problems (indeed, it seems that RK4 integrated over a slightly shorter total toroidal angle).

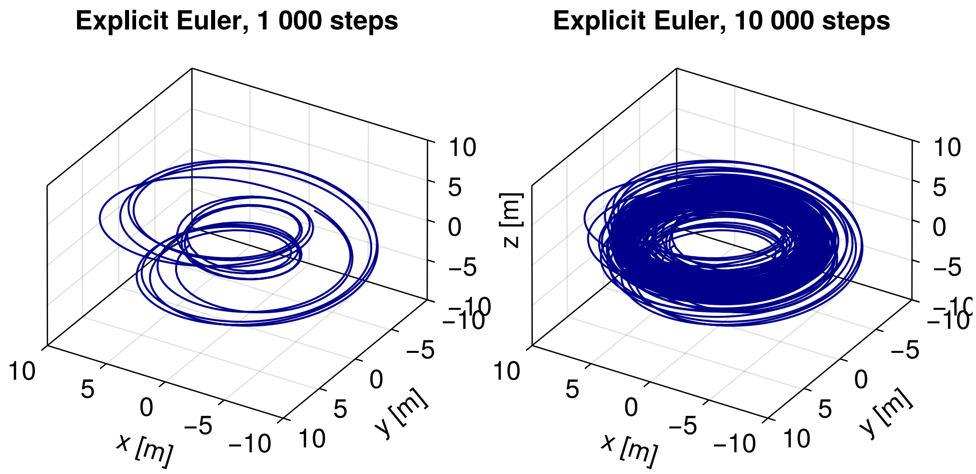

Figure 5.8 Magnetic field lines integrated by the Explicit Euler method. The parameters are the same as in Fig. 5.7. Left figure uses $N_{\text{steps}} = 10^4$, right figure uses $N_{\text{steps}} = 10^5$.

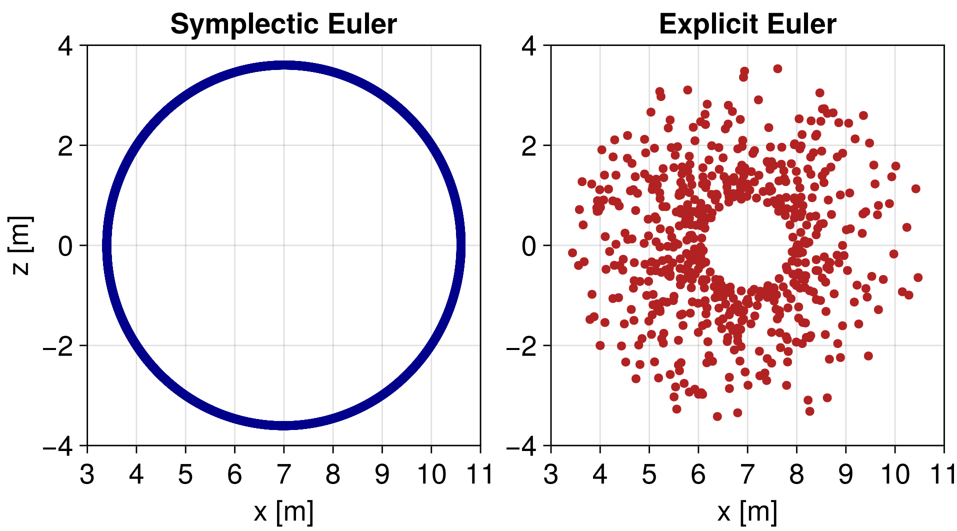

Figure 5.9 Poincaré map given by the intersection of the magnetic field lines with the xz plane. The Explicit and Symplectic Euler methods are compared. The physical parameters are the same as in Fig. 5.7. The number of steps is $N_{\text{steps}} = 5 \cdot 10^6$. The Symplectic Euler uses 3 FPI iterations and step size 0.01, whereas the Explicit Euler uses 0.001.

In Fig. 5.8, integration of the original problem (5.7) by the Explicit Euler method is shown. The emerging errors are clearly visible. Fig. 5.9 illustrates these errors even more using a *Poincaré map*, in which the intersections of the field line with a given plane are visualised. The Symplectic Euler method gives the correct result that the field line stays on the manifold given by a constant r coordinate (a torus surface), even though the Explicit Euler method uses a smaller step size.

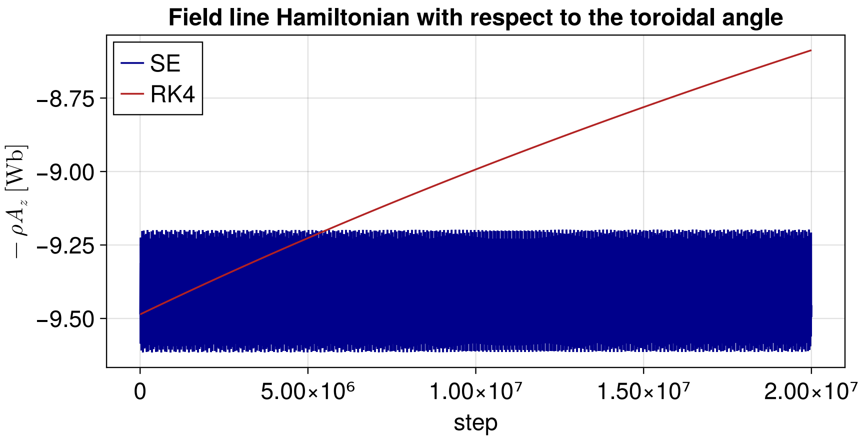

Figure 5.10 The evolution of the magnetic field line Hamiltonian with respect to the integration steps for the Symplectic Euler method and RK4. The parameters are $R = 7\,\mathrm{m}$, $B_0 = 5\,\mathrm{T}$, $a = 1$, $x_0 = 4\,\mathrm{m}$, $y_0 = 0\,\mathrm{m}$, $z_0 = -2\,\mathrm{m}$ (corresponding to $\rho_0 = 4\,\mathrm{m}$ for the Symplectic Euler method), $h = 0.05$, $N_{\text{steps}} = 2 \cdot 10^7$ and 4 iterations of the FPI method.

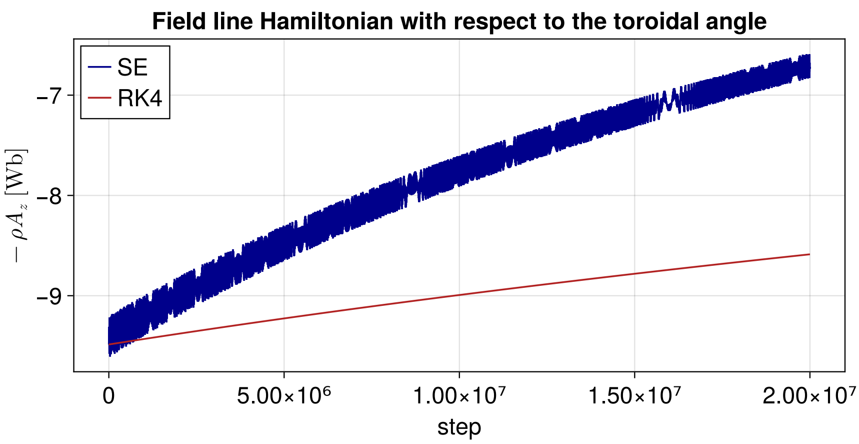

Figure 5.11 The evolution of the magnetic field line Hamiltonian with respect to the integration steps for the Symplectic Euler method and RK4. The parameters are the same as in Fig. 5.10, but only 3 iterations of the FPI method are used for the Symplectic Euler method.

In Fig. 5.10, the evolution of the Hamiltonian is shown both for the Symplectic Euler method (with 4 steps of FPI method) and for RK4. We chose a larger integration step $h = 0.05$ compared to previous examples, as RK4 calculates very precise approximations for smaller step sizes as a scheme of order 4. We see that this step size is insufficient for precise calculation by RK4, but might be sufficient for the Symplectic Euler method (or at least, the Hamiltonian stays approximately constant during the computation).

However, compare this result with Fig. 5.11, where we just decreased the number of FPI iterations to 3. We see that now the Symplectic Euler method does not conserve the Hamiltonian over this integration interval. Of course, good results cannot be expected when a constant number of FPI method iterations is used instead of a more sophisticated stopping criterion (and over a rather large integration interval), but the point is to demonstrate that implicit symplectic schemes do not always have the good properties of theoretical and explicit symplectic integrators.

Conclusion

In this thesis, we presented the class of symplectic integrators designed to solve Hamiltonian systems of ordinary differential equations. We demonstrated both theoretically and empirically that these numerical integrators have advantageous properties.

The limitations of this approach were also explored, particularly how the incorporation of a root-finding algorithm perturbs the matrix of symplectic structure, which should be invariant in an idealised scenario when a symplectic integrator is used. The absence of such results in existing literature is surprising and we believe that this topic should be more explored.

In particular, we are interested how the result of Theorem 10 could be generalised for other symplectic schemes and dimensions $N \geq 2$. Another relevant question is how to interpret the deviation blocks whose emergence was shown in Section 4.2.

For some Hamiltonians, it can be the case that one of the Symplectic Euler schemes is explicit and the other one is implicit. Such systems would be perfect candidates for empirical studies of the influence of symplectic perturbation to the performance of the integrator.

Given that the class of symplectic integrators is still relatively new (at least compared to the standard methods of numerical ODE integration), it is understandable that there are still unanswered questions about their behaviour. We are excited to see advances in this field.

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